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MIL-M-38510/343A
21 SEPTEMBER 1989
SUPERSEDING
MIL-M-38510/343
16 July 1986

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, BINARY COUNTERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, bipolar, Advanced Schottky TTL, binary counter microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete Part or Identifying Number (PIN) (see 6.7).

1.2 Classification.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit
01	Synchronous 4-bit binary counter (asynchronous master reset)
02	Synchronous 4-bit binary counter (synchronous reset)
03	Synchronous 4-bit up/down binary counter (with mode control)
04	Synchronous 4-bit up/down binary counter (asynchronous master reset)

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, .840" x .310" x .200"), dual-in-line package
F	F-5 (16-lead, .440" x .285" x .085"), flat package
X	C-2A (20 terminal, .358" x .358" x .075"), square chip carrier package
2	C-2 (20 terminal, .358" x .358" x .100"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441-5700, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5962
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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01, 02, 03, 04	303 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases E, F, X, 2	See MIL-M-38510, appendix C
Junction temperature (T_J) 2/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V dc minimum to 5.5 V dc maximum
Minimum high level input voltage (V_{IH})	2.0 V dc
Maximum low level input voltage (V_{IL})	0.8 V dc
Normalized fanout (each output) 3/:	
Low level	33 maximum
High level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Width of clock pulse, high ($\overline{PE}$ = High)	
Device types 01, 02	9.0 ns minimum
Width of clock pulse, high ($\overline{PE}$ = Low)	
Device types 01, 02	7.0 ns minimum
Width of clock pulse, low ($\overline{PE}$ = High)	
Device types 01, 02	8.0 ns minimum
Width of clock pulse, low ($\overline{PE}$ = Low)	
Device types 01, 02	9.0 ns minimum
Width of master reset pulse, low ($\overline{MR}$ = low)	
Device type 01	9.5 ns minimum
Width of $\overline{PL}$ pulse low:	
Device type 03	8.5 ns minimum
Width of $\overline{PL}$ pulse low:	
Device type 04	7.5 ns minimum
Width of clock pulse low	
Device type 03	7.0 ns minimum
Width of CPU or CPD pulse low	
Device type 04	7.0 ns minimum
Width of master reset pulse, high (MR = high)	
Device type 04	6.0 ns minimum
Width of CPU or CPD pulse, low (change of direction)	
Device type 04	12.0 ns minimum
Setup time Pn high to clock pulse	
Device types 01, 02	5.5 ns minimum
Setup time Pn low to clock pulse	
Device types 01, 02	5.5 ns minimum
Setup time $\overline{PE}$ or $\overline{SR}$ high to clock pulse	
Device types 01, 02	13.5 ns minimum
Setup time $\overline{PE}$ or $\overline{SR}$ low to clock pulse	
Device types 01, 02	10.5 ns minimum

1/ Must withstand added P_D due to short circuit test, e.g., I_{OS}.
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.
3/ The device shall fanout in both high and low levels to the specified number of inputs of the same device type as that being tested.

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Setup time CEP or CET high to clock pulse	
Device types 01, 02	13.0 ns minimum
Setup time CEP or CET low to clock pulse	
Device types 01, 02	7.5 ns minimum
Setup time $\overline{U}$/D high to clock pulse	
Device type 03	12.0 ns minimum
Setup time $\overline{U}$/D low to clock pulse	
Device type 03	12.0 ns minimum
Setup time Pn high to $\overline{PL}$	
Device types 03, 04	6.0 ns minimum
Setup time Pn low to $\overline{PL}$	
Device types 03, 04	6.0 ns minimum
Setup time $\overline{CE}$ low to clock pulse	
Device type 03	10.5 ns minimum
Hold time Pn high to clock pulse	
Device types 01, 02	2.5 ns minimum
Hold time Pn low to clock pulse	
Device types 01, 02	2.5 ns minimum
Hold time $\overline{PE}$ or $\overline{SR}$ high to clock pulse	
Device types 01, 02	2.0 ns minimum
Hold time $\overline{PE}$ or $\overline{SR}$ low to clock pulse	
Device types 01, 02	0.0 ns minimum
Hold time CEP or CET high to clock pulse	
Device types 01, 02	2.0 ns minimum
Hold time CEP or CET low to clock pulse	
Device types 01, 02	2.0 ns minimum
Hold time Pn low to $\overline{PL}$	
Device types 03, 04	2.0 ns minimum
Hold time $\overline{U}$/D high to clock pulse	
Device type 03	0.0 ns minimum
Hold time $\overline{U}$/D low to clock pulse	
Device type 03	0.0 ns minimum
Hold time $\overline{CE}$ low to clock pulse	
Device type 03	0.0 ns minimum
Recovery time master reset to clock pulse	
Device type 01	6.0 ns minimum
Recovery time $\overline{PL}$ to clock pulse	
Device type 03	7.5 ns minimum
Recovery time master reset to CPU or CPD	
Device type 04	4.5 ns minimum
Recovery time $\overline{PL}$ to CPU or CPD	
Device type 04	8.0 ns minimum

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this specification takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Logic diagram. The logic diagram shall be as specified on figure 2.

3.2.3 Truth table. The truth table shall be as specified on figure 3.

3.2.4 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics are as specified in table I, and apply over the full case operating temperature range.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits		Unit
				Min	Max	
High level output voltage	V_{OH}	V_{CC} = 4.5 V; V_{IL} = 0.8 V; I_{OH} = -1.0 mA; V_{IH} = 2.0 V	All	2.5		V
Low level output voltage	V_{OL}	V_{CC} = 4.5 V; V_{IH} = 2.0 V; I_{OL} = 20 mA; V_{IL} = 0.8 V	All		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V; I_{IN} = -18 mA; T_C = 25°C	All		-1.2	V
High level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IN} = 2.7 V	01,02		40	µA
			03,04		20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IN} = 7.0 V	All		100	µA
Low level input current	I_{IL1}	V_{CC} = 5.5 V; V_{IL} = 0.5 V	01,02	-0.0	-0.6	mA
			03,04	-.03	-0.6	mA
	I_{IL2}	V_{CC} = 5.5 V; V_{IL} = 0.5 V	01,02	-0.0	-1.2	mA
			03,04	-.09	-1.8	mA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V; V_{OS} = 0.0 V	All	-60	-150	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	All		55	mA
Maximum count frequency	f_{MAX}	V_{CC} = 5.0 V	All	70		MHz
Propagation delay time, CP to Qn	t_{PLH1}	V_{CC} = 5.0 V; C_L = 50 pF ±10% See figure 5	03	3.0	9.5	ns
CP to Qn	t_{PHL1}		03	5.0	13.5	ns
CPU, CPD to Qn	t_{PLH1}		04	3.5	10.0	ns
CPU, CPD to Qn	t_{PHL1}		04	5.5	14.0	ns
CP to Qn $\overline{PE}$ = (high)	t_{PLH1}		01,02	2.0	9.0	ns

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
CP to Qn $\overline{PE}$ = (high)	t_{PHL1}	V_{CC} = 5.0 V; C_L = 50 pF ±10% See figure 5	01,02	3.5	11.5	ns
CP to TC	t_{PLH2}		03	5.0	16.5	ns
CP to TC	t_{PHL2}		03	4.5	13.5	ns
CPU to $\overline{TCU}$	t_{PLH2}		04	4.0	10.5	ns
CPU to $\overline{TCU}$	t_{PHL2}		04	3.5	9.5	ns
$\overline{PL}$ to Qn	t_{PLH3}		04	5.0	13.5	ns
$\overline{PL}$ to Qn	t_{PHL3}		04	5.5	15.0	ns
CPD to $\overline{TCD}$	t_{PLH4}		04	4.0	10.5	ns
CPD to $\overline{TCD}$	t_{PHL4}		04	3.5	9.5	ns
CP to Qn $\overline{PE}$ = (Low)	t_{PLH2}		01,02	2.0	10.0	ns
CP to Qn $\overline{PE}$ = (Low)	t_{PHL2}		01,02	4.0	10.0	ns
CP to $\overline{RC}$	t_{PLH3}		03	3.0	11.5	ns
CP to $\overline{RC}$	t_{PHL3}		03	3.0	12.5	ns
CP to TC	t_{PLH3}		01,02	4.5	16.5	ns
CP to TC	t_{PHL3}		01,02	5.0	18.5	ns
Pn to Qn	t_{PLH4}		03	2.0	9.0	ns
Pn to Qn	t_{PHL4}		03	6.0	16.0	ns
Pn to Qn	t_{PLH5}		04	3.0	8.5	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
Pn to Qn	t_{PHL5}	V_{CC} = 5.0 V; C_L = 50 pF ±10% See figure 5	04	6.0	16.5	ns
CET to TC	t_{PLH4}		01,02	2.5	9.0	ns
CET to TC	t_{PHL4}		01,02	2.5	9.0	ns
$\overline{CE}$ to $\overline{RC}$	t_{PLH5}		03	3.0	9.0	ns
$\overline{CE}$ to $\overline{RC}$	t_{PHL5}		03	3.0	9.0	ns
$\overline{MR}$ to Qn	t_{PHL5}		01	5.5	14.0	ns
$\overline{MR}$ to TC	t_{PHL6}		01	4.5	14.0	ns
$\overline{PL}$ to Qn	t_{PLH6}		03	5.0	15.0	ns
$\overline{PL}$ to Qn	t_{PHL6}		03	5.5	14.5	ns
MR to $\overline{TCU}$	t_{PLH6}		04	5.0	15.0	ns
MR to $\overline{TCD}$	t_{PHL6}		04	6.0	16.0	ns
$\overline{U}$/D to $\overline{RC}$	t_{PLH7}		03	7.0	22.5	ns
$\overline{U}$/D to $\overline{RC}$	t_{PHL7}		03	5.5	14.0	ns
MR to Qn	t_{PHL11}		04	5.0	16.0	ns
$\overline{U}$/D to TC	t_{PLH8}		03	4.0	13.5	ns
$\overline{U}$/D to TC	t_{PHL8}		03	4.0	12.5	ns
$\overline{PL}$ to $\overline{TCU}$	t_{PLH7}		04	7.0	18.5	ns
$\overline{PL}$ to $\overline{TCU}$	t_{PHL7}		04	7.0	17.5	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
$\overline{PL}$ to $\overline{TCD}$	t_{PLH8}	V_{CC} = 5.0 V; C_L = 50 pF ±10% See figure 5	04	7.0	18.5	ns
$\overline{PL}$ to $\overline{TCD}$	t_{PHL8}		04	6.0	17.5	ns
Pn to $\overline{TCU}$	t_{PLH9}		04	7.0	16.5	ns
Pn to $\overline{TCU}$	t_{PHL9}		04	5.0	16.5	ns
Pn to $\overline{TCD}$	t_{PLH10}		04	5.5	16.5	ns
Pn to $\overline{TCD}$	t_{PHL10}		04	6.5	16.5	ns
MR to Qn	t_{PHL11}		04	5.0	16.0	ns

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 12 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical test parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A electrical test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9,10,11
Group B electrical test requirements (method 5005)	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test, method 1015 of MIL-STD-883.

(1) Test condition D, E, or F, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroups 7 and 8 tests shall verify the truth tables herein.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test conditions, method 1005 of MIL-STD-883.

(1) Test condition D, E, or F, using the circuit shown on figure 4, or equivalent.

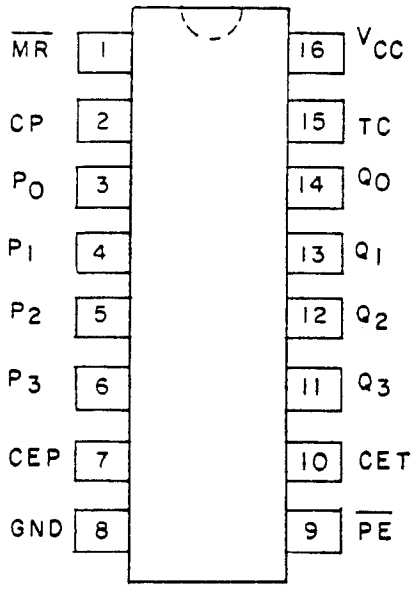
(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

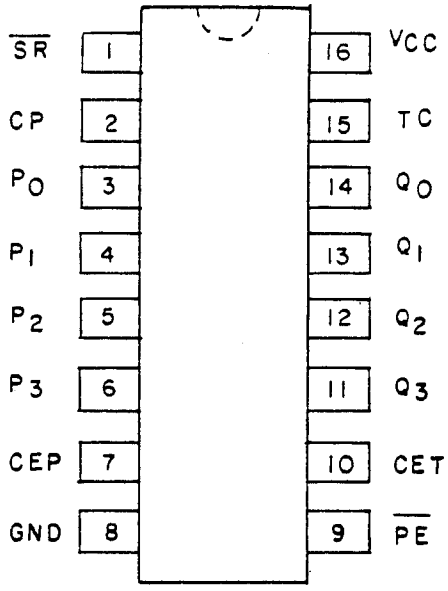
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Cases E and F

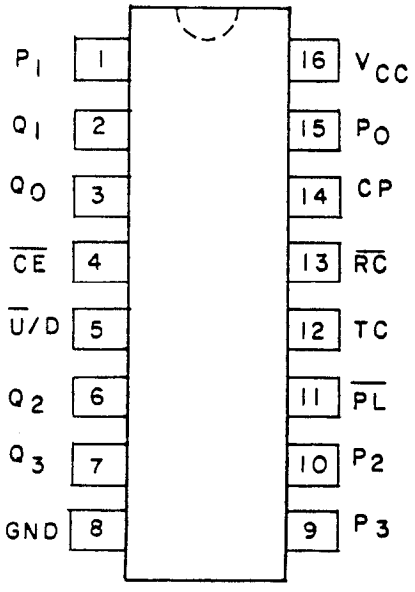
Device type 01



Device type 02



Device type 03



Device type 04

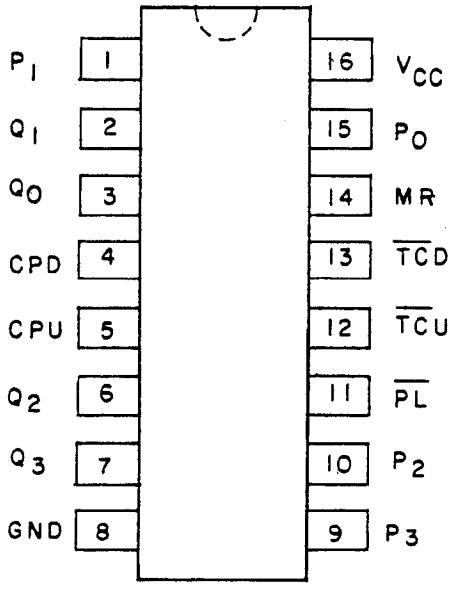


FIGURE 1. Terminal connections (top view).

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CASES 2 AND X

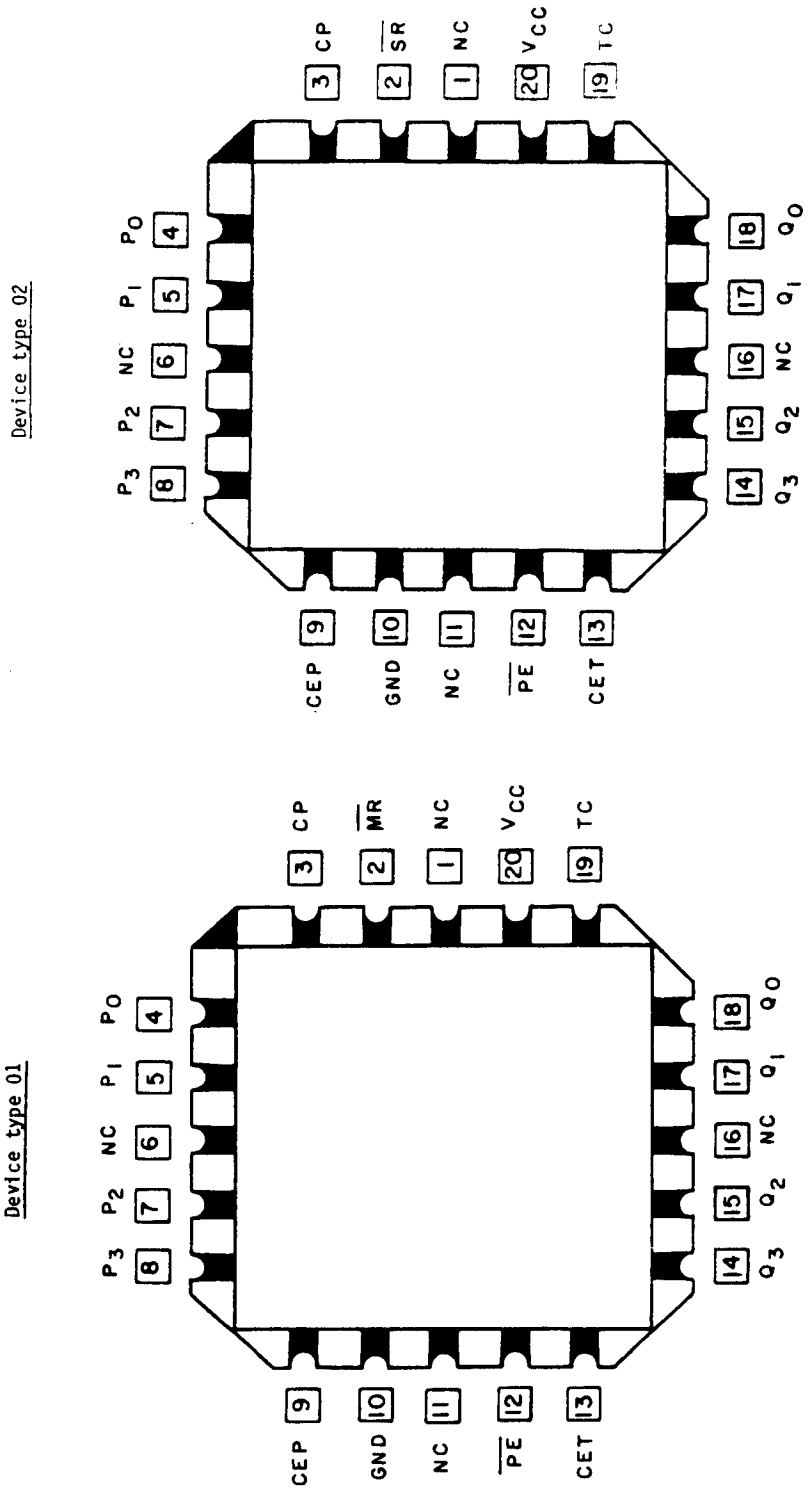


FIGURE 1. Terminal connections (top view) - Continued.

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CASES 2 AND X

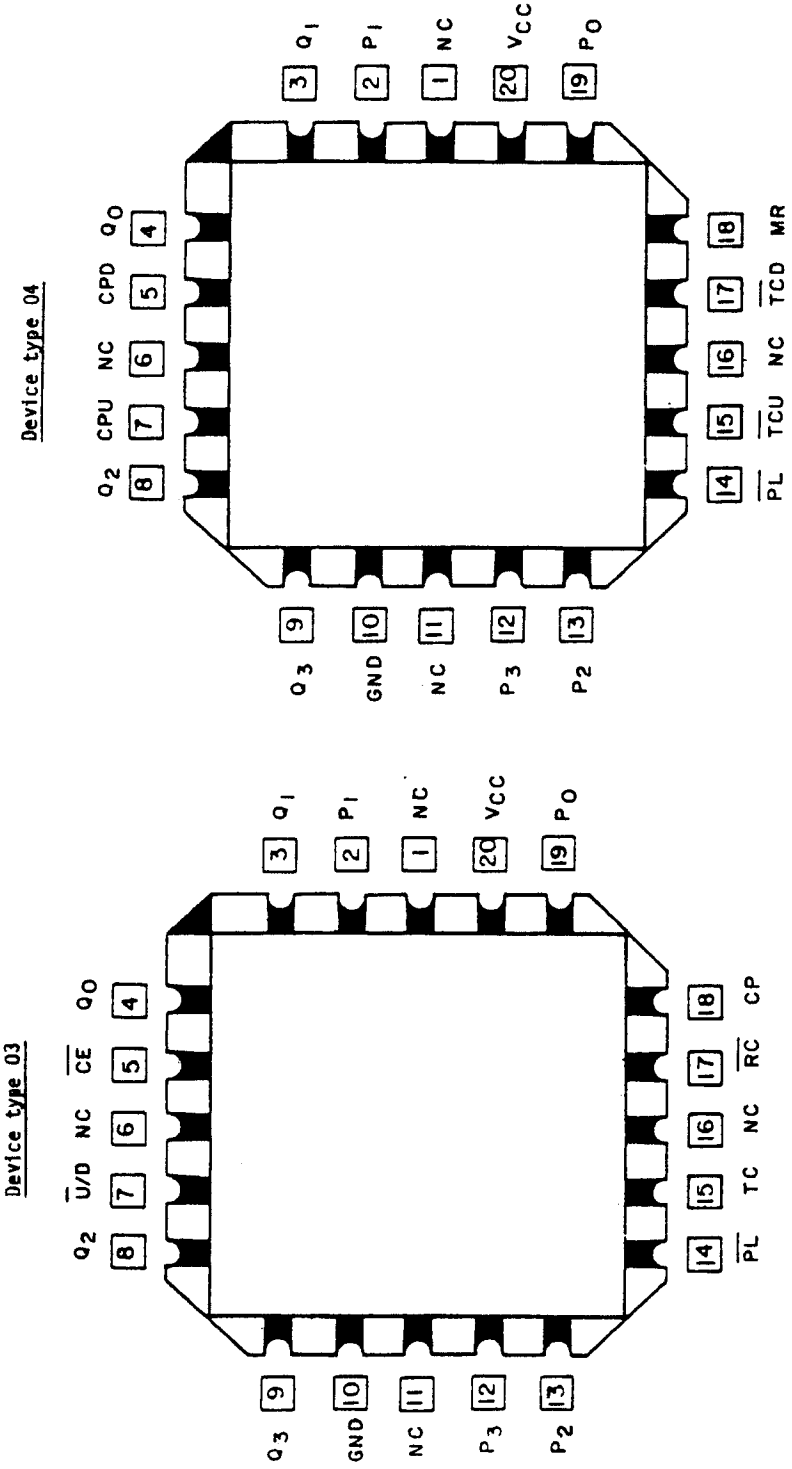


FIGURE 1. Terminal connections (top view) - Continued.

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Device types 01 and 02

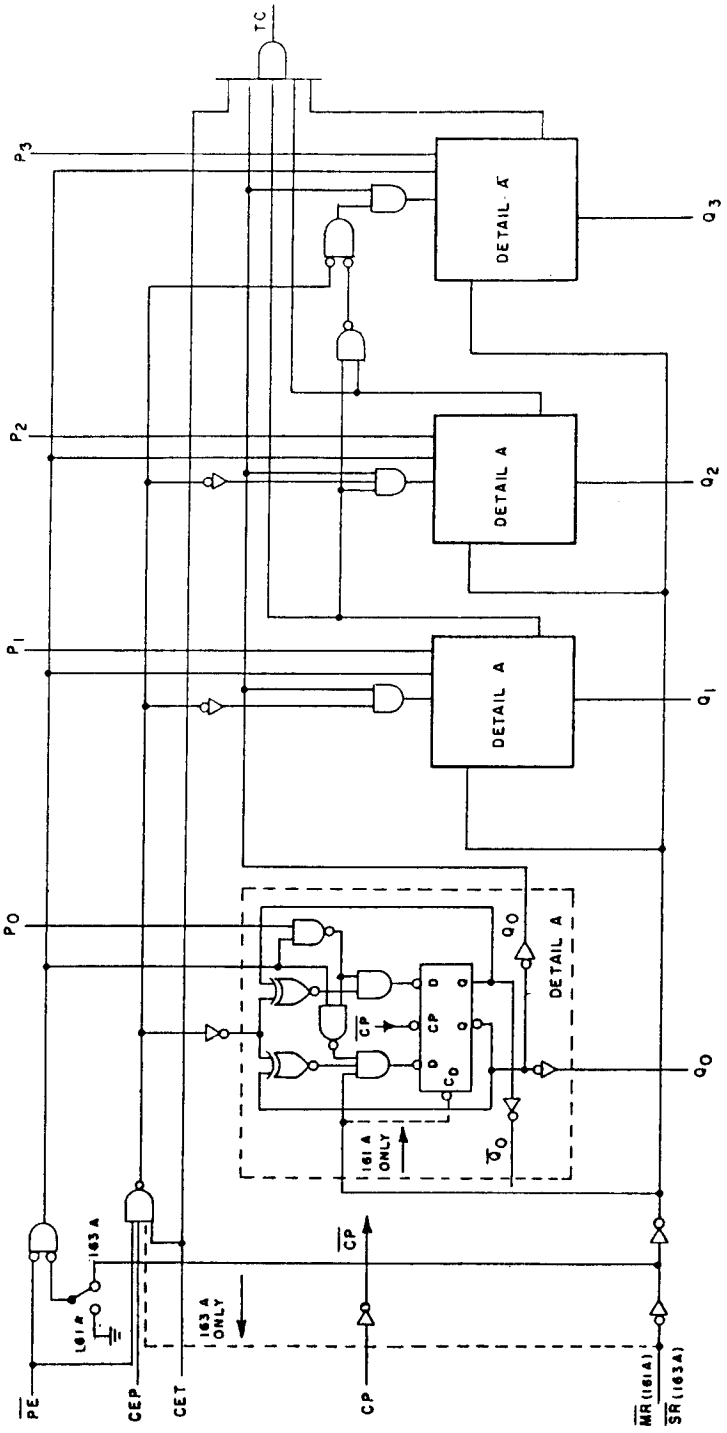


FIGURE 2. Logic diagram.

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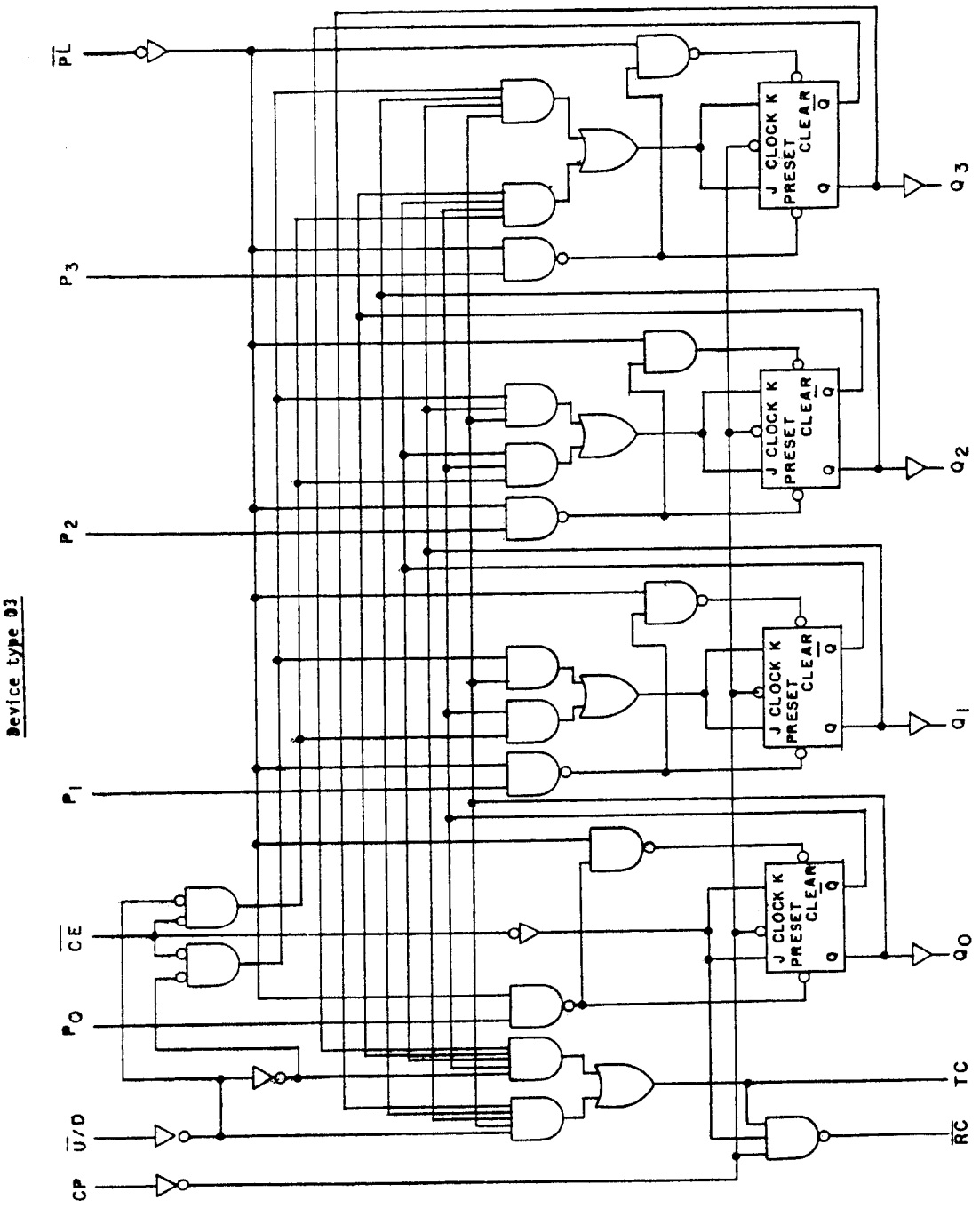


FIGURE 2. Logic diagram - Continued.

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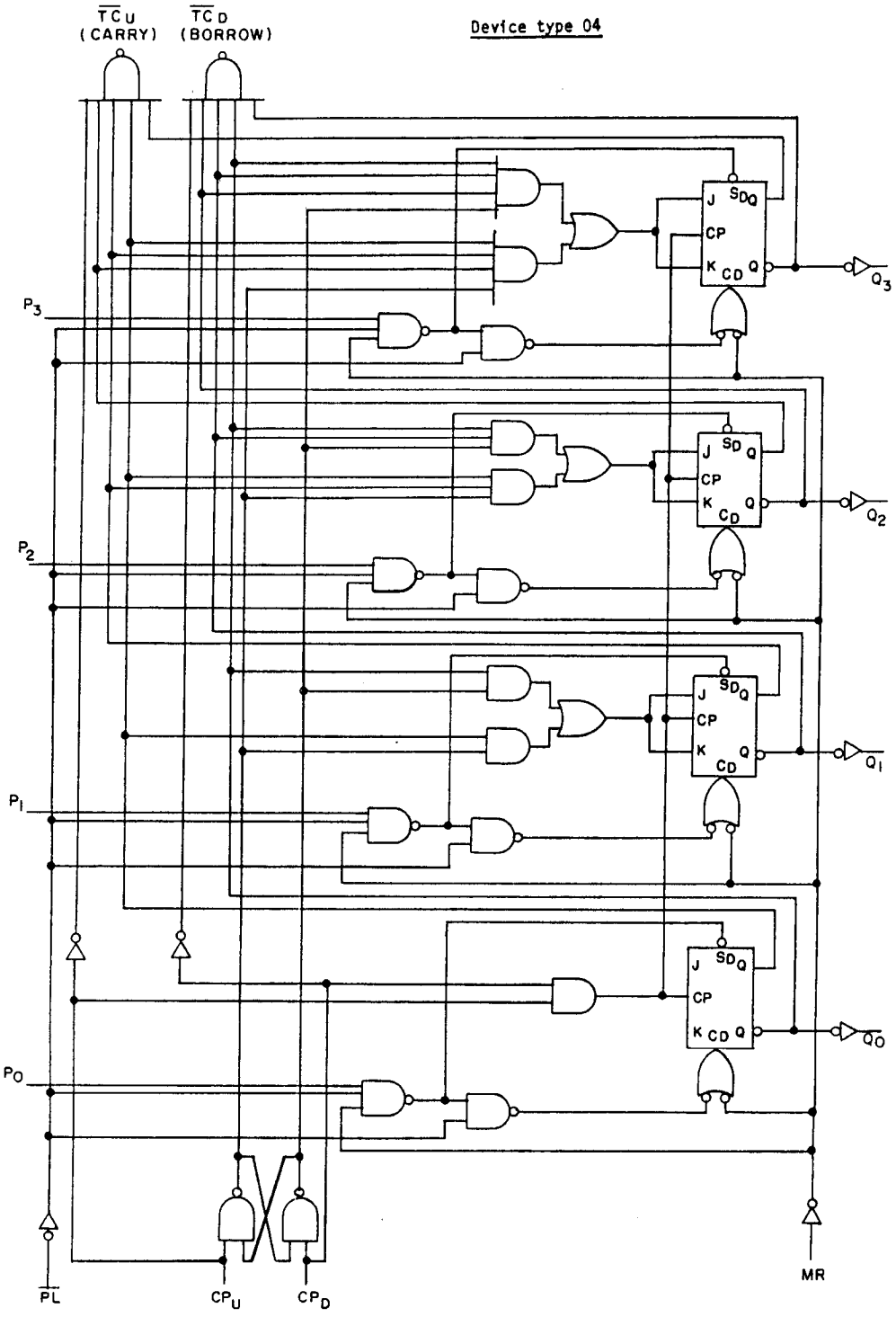


FIGURE 2. Logic diagram - Continued.

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Device types 01 and 02.

Mode select table

*$\overline{SR}$	$\overline{PE}$	CET	CEP	Action on the rising clock edge (⌐)
L	X	X	X	Reset (clear)
H	L	X	X	Load (P_n - Q_n)
H	H	H	H	Count (increment)
H	H	L	X	No change (hold)
H	H	X	L	No change (hold)

*For F163A only
H = High voltage level
L = Low voltage level
X = Immaterial

Device type 03

Mode select table

Inputs				Mode
$\overline{PL}$	$\overline{CE}$	U/D	CP	
H	L	L	⌐	Count up
H	L	H	⌐	Count down
L	X	X	X	Preset (asyn)
H	H	X	X	No change (hold)

RC truth table

Inputs			Output
$\overline{CE}$	TC*	CP	$\overline{RC}$
L	H	⌐⌐	⌐⌐
H	X	X	H
X	L	X	H

*TC is generated internally
H = High voltage level
L = Low voltage level
X = Immaterial
⌐ = Transition from low to high level
⌐⌐ = One low level pulse

Device type 04

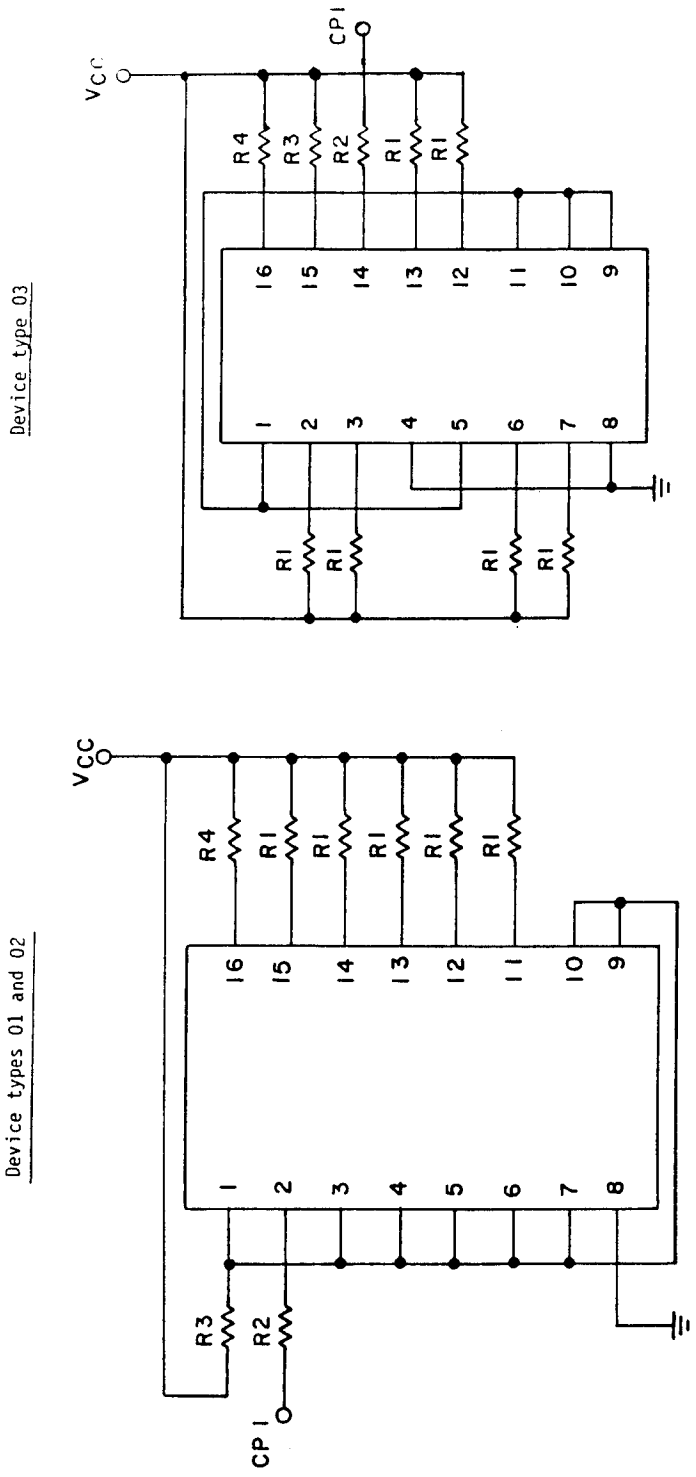
Function table

MR	$\overline{PL}$	CP_U	CP_D	Mode
H	X	X	X	Reset (asyn)
L	L	X	X	Preset (asyn)
L	H	H	H	No change
L	H	⌐	H	Count up
L	H	H	⌐	Count down

H = High voltage level
L = Low voltage level
X = Immaterial
⌐ = Transition from low to high level

FIGURE 3. Truth tables.

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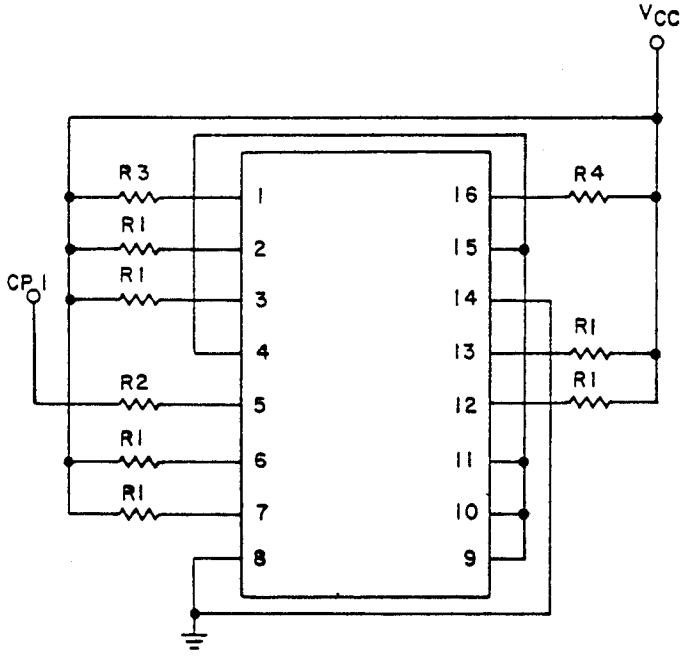


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%;
   V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω maximum; R2 = 51Ω ±5%; R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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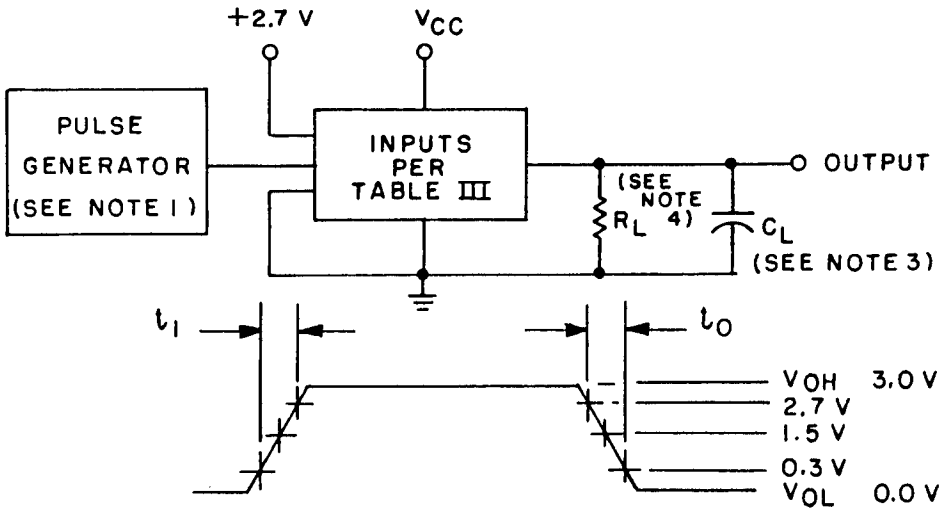
Device type 04



NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240 Ω maximum; R2 = 51Ω ±5% R3 = 1 kΩ maximum.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

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NOTES:
1. Pulse generator has the following characteristics: $t_I = t_O \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. Inputs not under test should be biased per table III.
3. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
4. R_L = 499Ω ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveform.

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Device types 01 and 02

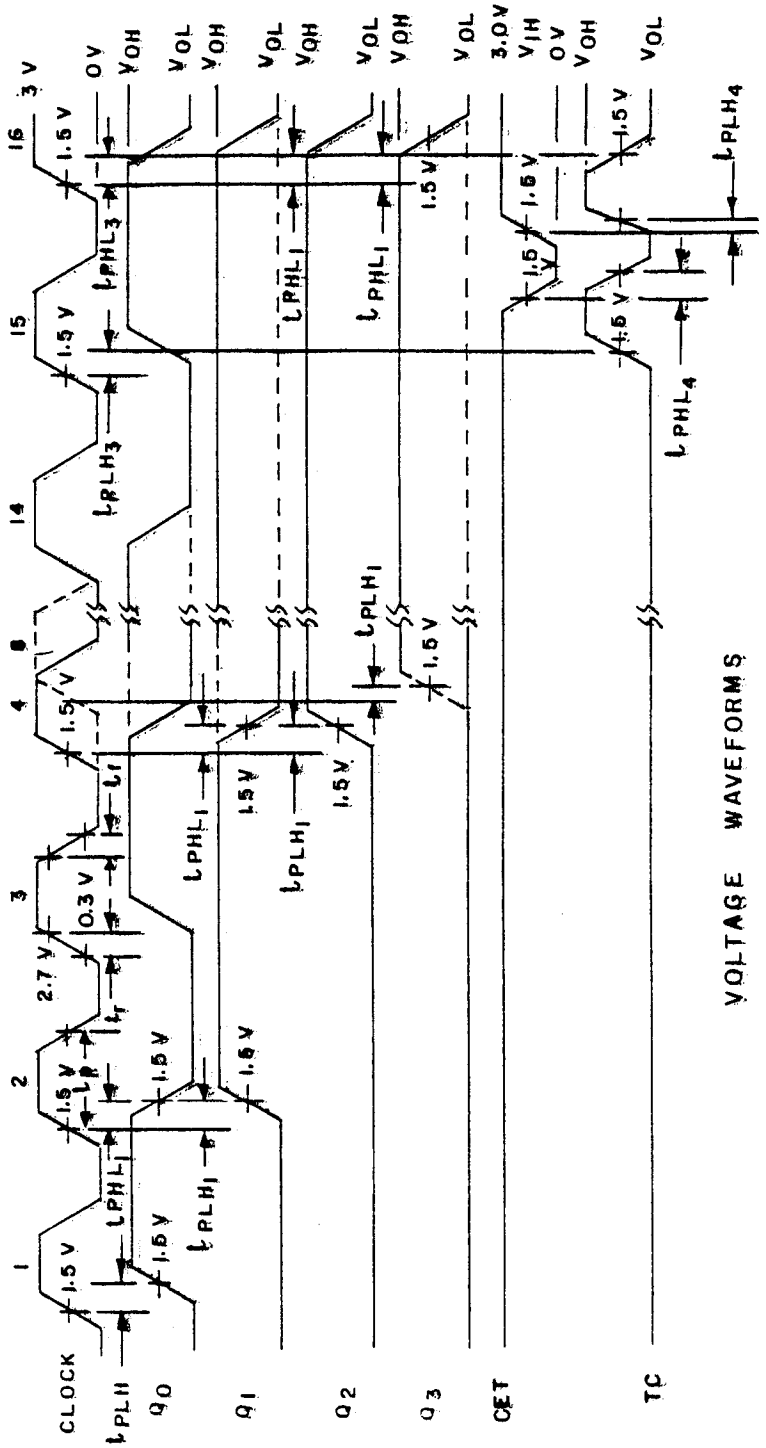
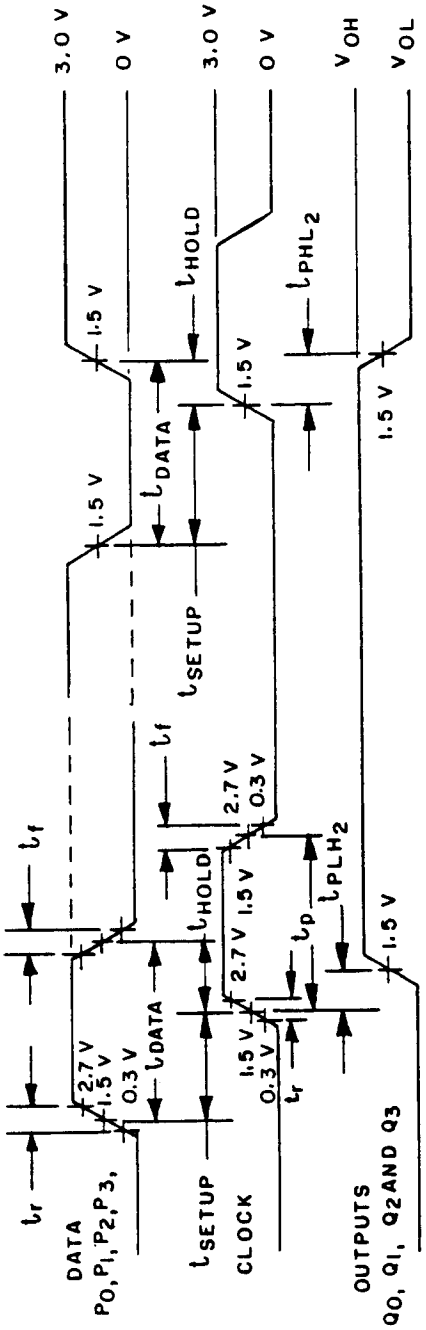


FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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NOTE: The data pulse generator has the following characteristics: V_{gen} = 3.0 V, $t_r \leq 2.5$ ns, $t_f \leq 2.5$ ns, t_{DATA} = 8.0 ns, t_{setup} = 5.5 ns, t_{HOLD} = 2.5 ns, t_p(clock) = 7.0 ns.

FIGURE 5. Switching time test circuit and waveforms for device types 01 and 02 - Continued.

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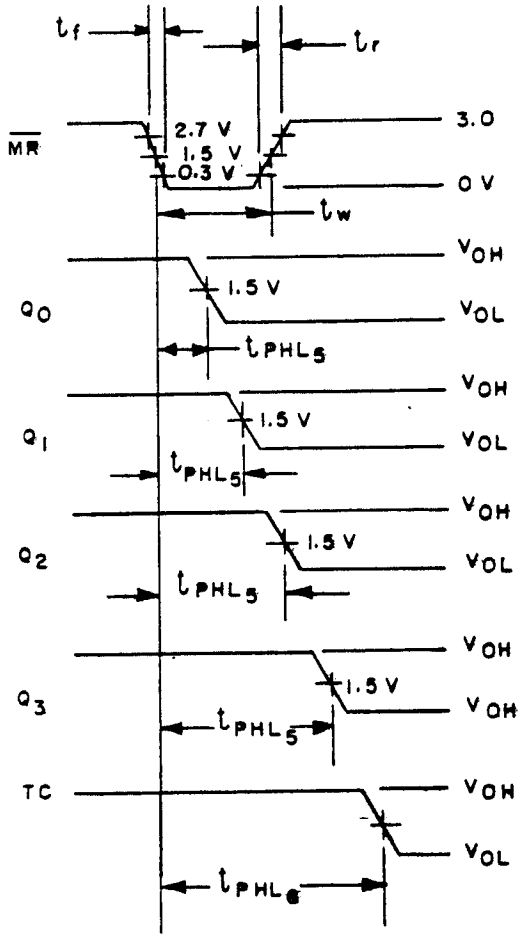
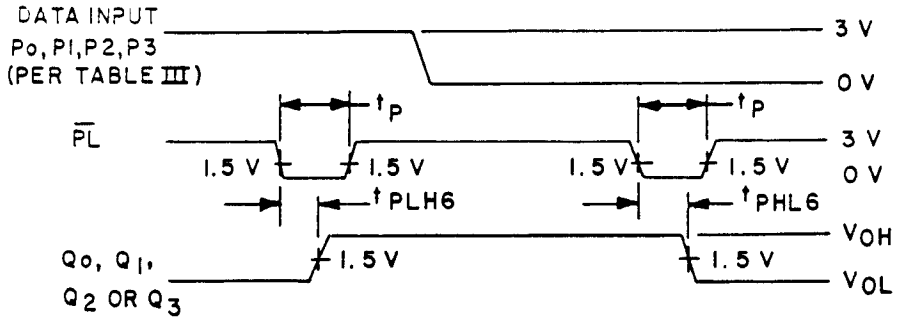
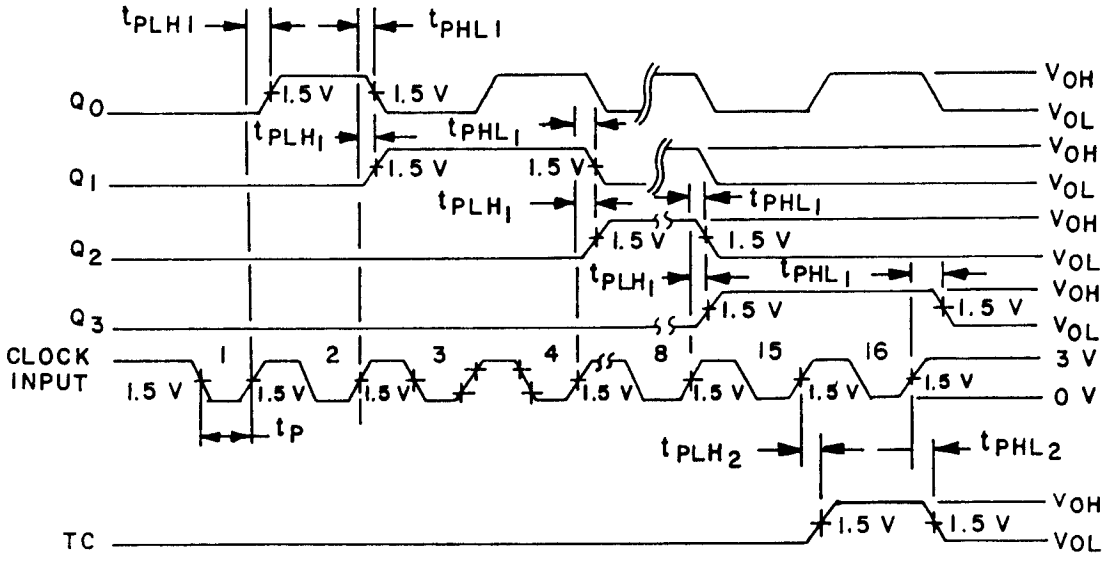


FIGURE 5. Switching time test circuit and waveforms for device type 01 - Continued.

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PARALLEL LOADED VOLTAGE WAVEFORMS

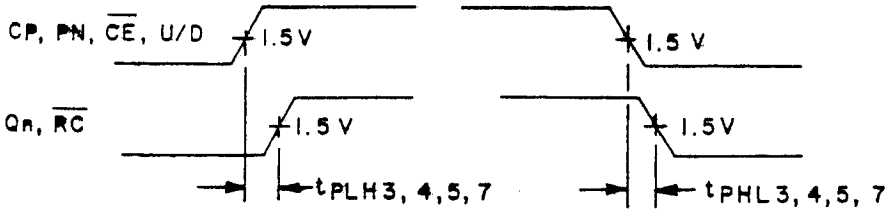


SERIAL LOADED VOLTAGE WAVEFORMS
(COUNT UP MODE)

FIGURE 5. Switching time test circuit and waveforms for device type 03 - Continued.

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Device type 03



Device type 03

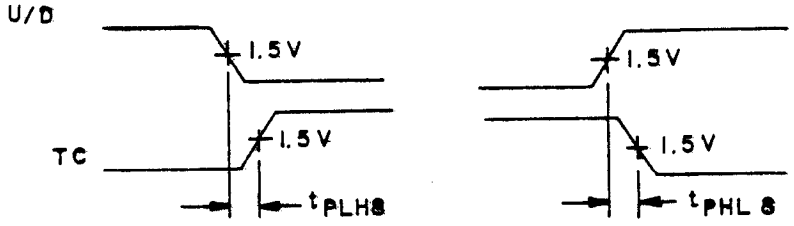
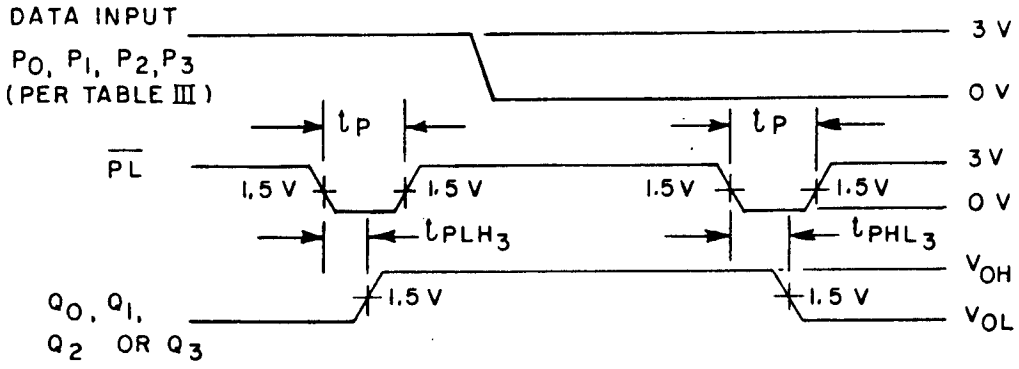
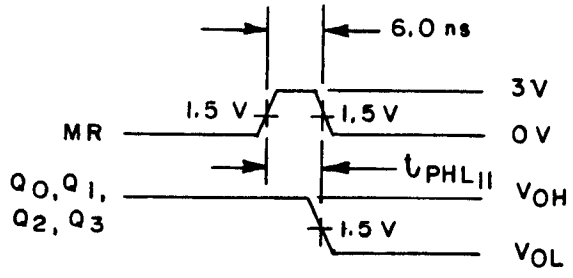


FIGURE 5. Switching time waveforms - Continued.

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PARALLEL LOADED VOLTAGE WAVEFORMS



CLEAR SWITCHING VOLTAGE WAVEFORM

FIGURE 5. Switching time test circuit and waveforms for device type 04 - Continued.

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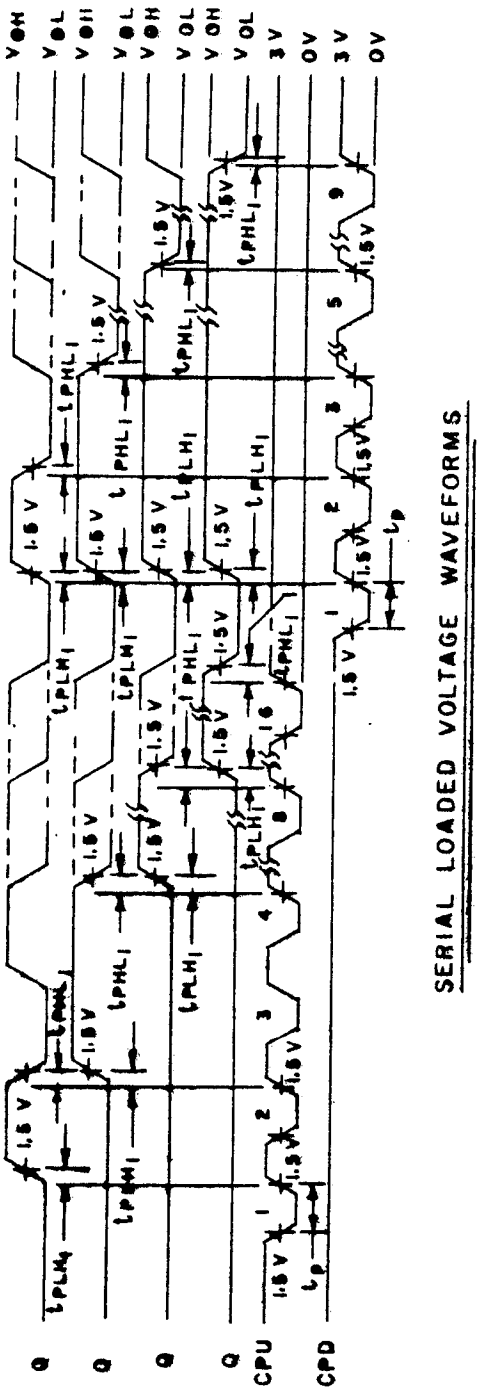
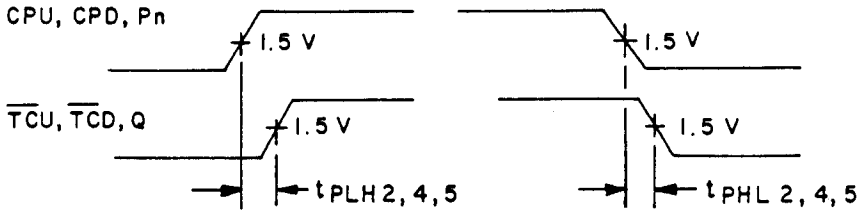


FIGURE 5. Switching time test circuit and waveforms for device type 04 - Continued.

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Device type 04



Device type 04

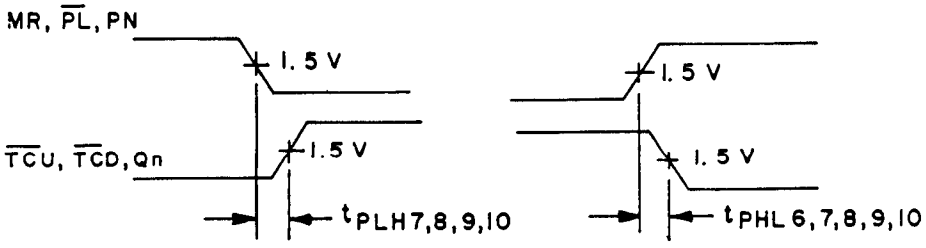


FIGURE 5. Switching time waveforms - Continued.

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TABLE III. Group A inspection for device 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / MR	2 / 3 / CP	3 / 4 / P_0	4 / 5 / P_1	5 / 7 / P_2	6 / 8 / P_3	7 / 9 / CEP	8 / 10 / GND	9 / 12 / PE	10 / 13 / CET	11 / 14 / Q_3	12 / 15 / Q_2	13 / 17 / Q_1	14 / 18 / Q_0	15 / 19 / TC	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5	2/ " " " "	2/ " " " "	2.0 V 5.5 V	2.0 V 5.5 V	2.0 V 5.5 V	2.0 V 5.5 V	5.5 V " " " "	GND " " " "	0.8 V " " " 0.0 V	5.5 V " " " 2.0 V	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	-1.0 mA	4.5 V " " " "	Q_3 Q_2 Q_1 Q_0 TC	2.5 " " " "		V " " " "
	V_{OL}	3007	6 7 8 9 10	5.5 V " " " "	" " " " "	0.8 V 5.5 V	0.8 V 5.5 V	0.8 V 5.5 V	0.8 V 5.5 V	" " " " "	" " " " "	" " " " "	5.5 V " " " 0.8 V	20 mA	20 mA	20 mA	20 mA	20 mA	" " " " "	Q_3 Q_2 Q_1 Q_0 TC		0.5 " " " "	" " " " "
	V_{IC}	3012	11 12 13 14 15 16 17 18 19	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	" " " " " " " " "	-18 mA	-18 mA						" " " " " " " " "	MR CP P_0 P_1 P_2 P_3 CEP PE CET		-1.2 " " " " " " " "	" " " " " " " " "
	I_{IH1}	3010	20 21 22 23 24 25 26 27 28	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	" " " " " " " " "	0.0 V 2.7 V 0.0 V	0.0 V 0.0 V 2.7 V						5.5 V " " " " " " " "	MR CP P_0 P_1 P_2 P_3 CEP PE CET		40 " " " " " " " "	µA " " " " " " " "
	I_{IH2}	3010	29 30 31 32 33 34 35 36 37	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	" " " " " " " " "	0.0 V 7.0 V 0.0 V	0.0 V 0.0 V 7.0 V						" " " " " " " " "	MR CP P_0 P_1 P_2 P_3 CEP PE CET		100 " " " " " " " "	" " " " " " " " "
	I_{IL1}	3009	38 39 40 41 42 43 44	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	" " " " " " "	0.0 V " " " 5.5 V	5.5 V						" " " " " " "	MR CP P_0 P_1 P_2 P_3 CEP	3/ " " " " " "	3/ " " " " " "	mA " " " " " "

See footnotes at end of table III.

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TABLE III. Group A inspection for device 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / $\overline{MR}$	2 / 3 / CP	3 / 4 / P_0	4 / 5 / P_1	5 / 7 / P_2	6 / 8 / P_3	7 / 9 / CEP	8 / 10 / GND	9 / 12 / $\overline{PE}$	10 / 13 / CET	11 / 14 / Q_3	12 / 15 / Q_2	13 / 17 / Q_1	14 / 18 / Q_0	15 / 19 / TC	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IL2}	3009	45 46 4/	5.5 V						5.5 V "	GND "	0.5 V 5.5 V	5.5 V 0.5 V						5.5 V "	$\overline{PE}$ CET	3/ 3/	3/ 3/	mA "
	I_{OS}	3011	47 48 49 50 51	2/ " " " "	2/ " " " "	 5.5 V 5.5 V	 5.5 V 5.5 V	 5.5 V 5.5 V	5.5 V 5.5 V	" " " " "	" " " " "	0.0 V " " " "	5.5 V " " " "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" " " " "	Q_3 Q_2 Q_1 Q_0 TC	-60 " " " "	-150 " " " "	" " " " "
	I_{CC}	3005	52	5.5 V	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	"	"	5.5 V	"						"	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional 5/	3014	53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80	B A "	B B A B A B A B A B A B A B A A B A B A B A B A B A B A B A	B "	B "	B "	B "	B "	B A "	GND "	A "	B A "	L "	L "	L "	L L H H L L H H L L H H L L H H L L H H L L H H L L H H L	L "	6/ "	All outputs "			

See footnotes at end of table III.

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TABLE III. Group A inspection for device 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / $\overline{MR}$	2 / 3 / CP	3 / 4 / P_0	4 / 5 / P_1	5 / 7 / P_2	6 / 8 / P_3	7 / 9 / CEP	8 / 10 / GND	9 / 12 / $\overline{PE}$	10 / 13 / CET	11 / 14 / Q_3	12 / 15 / Q_2	13 / 17 / Q_1	14 / 18 / Q_0	15 / 19 / TC	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional 5/	3014	81	A	B	B	B	B	B	A	GND	A	A	H	H	L	L	L	6/	All outputs			
			82	"	A	"	"	"	"	"	"	"	"	"	"	"	H	H	"				
			83	"	B	"	"	"	"	"	"	"	"	"	"	"	L	L	"				
			84	"	A	"	"	"	"	"	"	"	"	"	"	H	L	"	"				
			85	"	B	"	"	"	"	"	"	"	"	"	"	"	L	H	"				
			86	"	A	"	"	"	"	"	"	"	"	"	"	"	H	H	"				
			87	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"				
			88	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			89	"	A	"	"	"	"	"	"	"	"	"	"	"	"	L	"				
			90	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"				
			91	"	A	"	"	"	"	"	"	"	"	"	"	"	"	H	"				
			92	"	B	"	"	"	"	"	"	"	A	"	"	"	"	H	"				
			93	"	B	"	"	"	"	"	"	"	"	"	"	"	"	L	"				
			94	"	A	"	"	"	"	"	"	"	"	L	L	L	L	"	"				
			95	"	A	"	"	"	"	"	"	"	B	"	"	"	"	"	"				
			96	"	B	A	A	A	A	"	"	"	"	"	"	"	"	"	"				
			97	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"				
			98	"	B	"	"	"	"	"	"	"	"	H	H	H	H	H	"				
			99	B	"	"	"	"	"	"	"	"	"	L	L	L	L	L	"				
			100	A	"	"	"	"	"	"	"	"	"	L	L	L	L	L	"				
			101	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"				
			102	"	B	B	B	B	B	B	"	"	"	H	H	H	H	H	"				
			103	"	A	B	B	B	B	B	"	"	"	L	L	H	L	L	"				
			104	"	B	A	A	A	A	A	"	"	"	L	L	H	L	L	"				
			105	"	A	A	A	A	A	A	"	"	"	L	L	H	H	L	"				
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C See fig. 5	f_{MAX} 7/	3003	106	2.7 V	IN					2.7 V	GND	2.7 V	2.7 V	OUT	OUT	OUT	OUT		5.0 V	Q_0 Q_1 Q_2 Q_3	90		MHz
			107	"	"					"	"	"	"						"		"		"
			108	"	"					"	"	"	"						"		"		"
			109	"	"					"	"	"	"						"		"		"
	t_{PLH1}	3003	110 8/	2/	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	2.0	7.5	ns
			111 "	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			112 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			113 "	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PHL1}	3003	114 "	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	3.5	10.0	"
			115 "	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			116 "	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			117 "	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PLH2}	3003	118	"	"					"	"	0.0 V	"	OUT	OUT	OUT	OUT		"	CP to Q_0	2.5	8.5	"
			119	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			120	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			121	"	"					"	"	"	"						"	CP to Q_3	"	"	"
	t_{PHL2}	3003	122	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0	4.0	"	"
			123	"	"					"	"	"	"						"	CP to Q_1	"	"	"
			124	"	"					"	"	"	"						"	CP to Q_2	"	"	"
			125	"	"					"	"	"	"						"	CP to Q_3	"	"	"

See footnotes at end of table III.

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TABLE III. Group A inspection for device 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{MR}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
9 T_C = +25°C See fig. 5	t_{PLH3}	3003	126	0.0 V	IN	0.0 V	2.7 V	2.7 V	2.7 V	2.7 V	GND	0.0 V	2.7 V					OUT	5.0 V	CP to TC	4.5	14.0	ns
	t_{PHL3}	3003	127	2/	IN	2.7 V	"	"	"	"	"	2.7 V	2.7 V					"	"	CP to TC	5.0	16.0	"
	t_{PLH4}	3003	128	2.7 V	0.0 V	"	"	"	"	"	"	0.0 V	IN					"	"	CET to TC	2.5	7.5	"
	t_{PHL4}	3003	129	2.7 V	2.7 V	"	"	"	"	"	"	0.0 V	IN					"	"	CET to TC	2.5	7.5	"
	t_{PHL5}	3003	130 131 132 133	IN " " "	0.0 V " " "	"	"	"	"	" " " "	" " " "	0.0 V " " "	2.7 V " " "	OUT	OUT	OUT	OUT		" " " "	$\overline{MR}$ to Q_0 $\overline{MR}$ to Q_1 $\overline{MR}$ to Q_2 $\overline{MR}$ to Q_3	5.5 " " "	12.0 " " "	" " " "
	t_{PHL6}	3003	134	"	"	"	"	"	"	"	"	"	"					OUT	"	$\overline{MR}$ to TC	4.5	11.5	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of table III.

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TABLE III. Group A inspection for device 02.
Terminal conditions (pins not designated may be high $\geq$ 2.0 V, low $\leq$ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / $\overline{SR}$	2 / 3 / CP	3 / 4 / P_0	4 / 5 / P_1	5 / 7 / P_2	6 / 8 / P_3	7 / 9 / CEP	8 / 10 / GND	9 / 12 / $\overline{PE}$	10 / 13 / CET	11 / 14 / Q_3	12 / 15 / Q_2	13 / 17 / Q_1	14 / 18 / Q_0	15 / 19 / TC	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2/	2.0 V				0.8 V	GND	0.8 V			-1.0 mA		-1.0 mA		4.5 V	Q_0	2.5		V
			2	5.5 V	"		2.0 V				"	"							"	Q_1	"		"
			3	"	"			2.0 V			"	"							"	Q_2	"		"
			4	"	"				2.0 V		"	"		-1.0 mA					"	Q_3	"		"
			5	"	"	5.5 V	5.5 V	5.5 V	5.5 V		"	0.0 V	2.0 V					-1.0 mA	"	TC	"		"
	V_{OL}	3007	6	"	"	0.8 V					"	0.0 V				20 mA			"	Q_0		0.5	"
			7	"	"		0.8 V				"	"			20 mA				"	Q_1		"	"
			8	"	"			0.8 V			"	"					20 mA		"	Q_2		"	"
			9	"	"				0.8 V		"	"		20 mA					"	Q_3		"	"
			10	"	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	0.8 V					20 mA	"	TC		"	"
	V_{IC}	3022	11	-18 mA	-18 mA						"								"	$\overline{SR}$		-1.2	"
			12			-18 mA					"								"	CP		"	"
			13				-18 mA				"								"	P_0		"	"
			14					-18 mA			"								"	P_1		"	"
			15						-18 mA		"								"	P_2		"	"
			16							-18 mA	"								"	P_3		"	"
			17								"								"	CEP		"	"
			18								"	-18 mA							"	$\overline{PE}$		"	"
			19								"		-18 mA						"	CET		"	"
	I_{IH1}	3010	20	2.7 V	2.7 V						"								5.5 V	$\overline{SR}$		40	µA
			21			2.7 V					"								"	CP		20	"
			22				2.7 V				"								"	P_0		"	"
			23					2.7 V			"								"	P_1		"	"
			24						2.7 V		"								"	P_2		"	"
			25							2.7 V	"	0.0 V	0.0 V						"	P_3		"	"
			26							0.0 V	"	2.7 V	0.0 V						"	CEP		"	"
			27							0.0 V	"	0.0 V	0.0 V						"	$\overline{PE}$		40	"
			28		2/						"		2.7 V						"	CET		40	"
	I_{IH2}	3010	29	7.0 V							"								"	$\overline{SR}$		100	"
			30		7.0 V						"								"	CP		"	"
			31			7.0 V					"								"	P_0		"	"
			32				7.0 V				"								"	P_1		"	"
			33					7.0 V			"								"	P_2		"	"
			34						7.0 V		"								"	P_3		"	"
			35							7.0 V	"	0.0 V	0.0 V						"	CEP		"	"
			36							0.0 V	"	7.0 V	0.0 V						"	$\overline{PE}$		"	"
			37		2/					0.0 V	"	0.0 V	7.0 V						"	CET		"	"
	I_{IL1}	3009	38		0.5 V						"								"	CP	3/	3/	mA
			39			0.5 V					"								"	P_0	"	"	"
			40				0.5 V				"								"	P_1	"	"	"
			41					0.5 V			"								"	P_2	"	"	"
			42						0.5 V		"								"	P_3	"	"	"
			43							0.5 V	"	5.5 V	5.5 V						"	CEP	"	"	"
			~~44~~	~~0.5 V~~						~~5.5 V~~	"	~~5.5 V~~	~~5.5 V~~						"	~~$\overline{SR}$~~	"	"	"

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TABLE III. Group A inspection for device 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / $\overline{SR}$	2 / 3 / CP	3 / 4 / P_0	4 / 5 / P_1	5 / 7 / P_2	6 / 8 / P_3	7 / 9 / CEP	8 / 10 / GND	9 / 12 / $\overline{PE}$	10 / 13 / CET	11 / 14 / Q_3	12 / 15 / Q_2	13 / 17 / Q_1	14 / 18 / Q_0	15 / 19 / TC	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{IL2}	3009	~~45~~ 45 46 4/	0.5 V 5.5 V	 2/	 5.5 V	 5.5 V	 5.5 V	 5.5 V	5.5 V 5.5 V	GND "	0.5 V 2/	5.5 V 0.5 V						5.5 V "	$\overline{SR}$ PE CET	3/ 3/	3/ 3/	mA "
	I_{OS}	3011	47 48 49 50 51	5.5 V " " " "	2/ " " " "	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	5.5 V 5.5 V	 5.5 V	" " " " "	0.0 V " " " "	 5.5 V	 0.0 V	 0.0 V	 0.0 V	0.0 V	 0.0 V	" " " " "	O_0 O_1 O_2 O_3 TC	-60 " " " "	-150 " " " "	" " " " "
	I_{CC}	3005	52	5.5 V	5.5 V	0.0 V	0.0 V	0.0 V	0.0 V	5.5 V	"	5.5 V	5.5 V						5.5 V	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional 5/	3014	53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83	B A "	2/ B A B A B A B A B A B A B A A B A B A B A B A B A B A B A B	B "	B "	B "	B "	B "	B A "	GND "	A "	B A "	L "	L "	L "	L L H H L L H H L L H H L L H H L L H H L L H H L L H H L L H H	L "	6/ "	All outputs "			

See footnotes at end of table III.

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TABLE III. Group A inspection for device 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
			Test no.	$\overline{SR}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}				
7 T_C = +25°C	Functional 5/	3014	84	A	A	B	B	B	B	A	GND	A	A	H	H	H	L	L	6/	All outputs			
			85	"	B	"	"	"	"	"	"	"	"	"	"	"	L	L	"	"			
			86	"	A	"	"	"	"	"	"	"	"	"	"	"	H	H	"	"			
			87	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			88	"	B	"	"	"	"	"	"	"	B	"	"	"	"	"	"	"			
			89	"	A	"	"	"	"	"	"	"	"	"	"	"	"	L	"	"			
			90	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			91	"	A	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			92	"	B	"	"	"	"	"	"	"	A	"	"	"	"	H	"	"			
			93	"	B	"	"	"	"	"	"	"	"	"	"	"	"	H	"	"			
			94	"	A	"	"	"	"	"	"	"	"	L	L	L	L	L	"	"			
			95	"	A	"	"	"	"	"	"	B	"	"	"	"	"	"	"	"			
			96	"	B	A	A	A	A	"	"	"	"	"	"	"	"	"	"	"			
			97	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"	"			
			98	"	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			99	B	B	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"			
			100	B	A	"	"	"	"	"	"	"	"	L	L	L	L	L	"	"			
			101	A	B	"	"	"	"	"	"	"	"	L	L	L	L	L	"	"			
			102	"	A	"	"	"	"	"	"	"	"	H	H	H	H	H	"	"			
			103	"	B	B	B	B	B	B	"	"	"	H	H	H	H	H	"	"			
			104	"	A	B	B	B	B	B	"	"	"	L	L	L	L	L	"	"			
			105	"	B	A	A	A	A	A	"	"	"	L	L	L	L	L	"	"			
			106	"	A	A	A	A	A	A	"	"	"	H	H	H	H	H	"	"			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C See fig. 5	f_{MAX} 7/	3003	107	2.7 V	IN					2.7 V	GND	2.7 V	2.7 V	OUT	OUT	OUT	OUT		5.0 V	Q_0 Q_1 Q_2 Q_3	90		MHz
			108	"	"					"	"	"	"						"		"		"
			109	"	"					"	"	"	"						"		"		"
			110	"	"					"	"	"	"						"		"		"
	t_{PLH1}	3003	111 8/	2/	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	2.0	7.5	ns
			112	"	"					"	"	"	"						"		"		"
			113	"	"					"	"	"	"						"		"		"
			114	"	"					"	"	"	"						"		"		"
	t_{PHL1}	3003	115	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	3.5	10.0	"
			116	"	"					"	"	"	"						"		"		"
			117	"	"					"	"	"	"						"		"		"
			118	"	"					"	"	"	"						"		"		"
	t_{PLH2}	3003	119	2.7 V	"					"	"	0.0 V	0.0 V	OUT	OUT	OUT	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	2.5	8.5	"
			120	"	"					"	"	"	"						"		"		"
			121	"	"					"	"	"	"						"		"		"
			122	"	"					"	"	"	"						"		"		"
	t_{PHL2}	3003	123	"	"					"	"	"	"	OUT	OUT	OUT	OUT		"	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	4.0	"	"
			124	"	"					"	"	"	"						"		"		"
			125	"	"					"	"	"	"						"		"		"
			126	"	"					"	"	"	"						"		"		"

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TABLE III. Group A inspection for device 02 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases X,2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	$\overline{SR}$	CP	P_0	P_1	P_2	P_3	CEP	GND	$\overline{PE}$	CET	Q_3	Q_2	Q_1	Q_0	TC	V_{CC}		Min	Max	
9 T_C = +25°C See fig. 5	t_{PLH3}	3003	127	2.7 V	IN	2/	2.7 V	2.7 V	2.7 V	2.7 V	GND	2/	2.7 V					OUT	5.0 V	CP to TC	4.5	14.0	ns
	t_{PHL3}	3003	128	"	IN	2.7 V	"	"	"	"	"	2.7 V	2.7 V					"	"	CP to TC	5.0	16.0	"
	t_{PLH4}	3003	129	"	0.0 V	"	"	"	"	"	"	0.0 V	IN					"	"	CET to TC	2.5	7.5	"
	t_{PHL4}	3003	130	"	0.0 V	"	"	"	"	"	"	0.0 V	IN					"	"	CET to TC	2.5	7.5	"
10	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

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TABLE III. Group A inspection for device 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases X, 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	P_1	Q_1	Q_0	$\overline{CE}$	U/D	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	TC	$\overline{RC}$	CP	P_0	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	-1.0 mA		2.0 V				GND			0.8 V					4.5 V	Q_1	2.5		V
			2				"				"			"				2.0 V	"	Q_0	"		"
			3				"				"			"					"	Q_2	"		"
			4			-1.0 mA			-1.0 mA		"	2.0 V	2.0 V	"					"	Q_3	"		"
			5	2.0 V			0.8 V	0.8 V		-1.0 mA	"	"	2.0 V	"	-1.0 mA		2.0 V	2.0 V	"	$\overline{TC}$	"		"
			6	2.0 V			0.8 V	0.8 V			"	"	2.0 V	"		-1.0 mA	2.0 V	2.0 V	"	$\overline{RC}$	"		"
	V_{OL}	3007	7	0.8 V	20 mA		2.0 V				"			"					"	Q_1		0.5	"
			8				"				"			"					"	Q_0		"	"
			9				"				"			"					"	Q_2		"	"
			10				"				"			"					"	Q_3		"	"
			11	2.0 V			0.8 V	2.0 V			"	0.8 V	0.8 V	"	20 mA		0.8 V	0.8 V	"	$\overline{TC}$		"	"
			12	2.0 V				0.8 V			"	2.0 V	2.0 V	"		20 mA	0.8 V	2.0 V	"	$\overline{RC}$		"	"
	V_{IC}	3022	13	-18 mA																P_1		-1.2	
			14				-18 mA													$\overline{CE}$		"	
			15					-18 mA												U/D		"	
			16									-18 mA								P_3		"	
			17										-18 mA							P_2		"	
			18											-18 mA			-18 mA			$\overline{PL}$		"	
			19															-18 mA		CP		"	
			20																	P_0		"	
	I_{IH1}	3010	21	2.7 V										5.5 V					5.5 V	P_1		20	µA
			22				2.7 V							"					"	$\overline{CE}$		"	"
			23					2.7 V						"					"	U/D		"	"
			24									2.7 V		5.5 V					"	P_3		"	"
			25										2.7 V	5.5 V					"	P_2		"	"
			26											2.7 V					"	$\overline{PL}$		"	"
			27														2.7 V		"	CP		"	"
			28											5.5 V				2.7 V	"	P_0		"	"
	I_{IH2}	3010	29	7.0 V										5.5 V					"	P_1		100	"
			30				7.0 V							"					"	$\overline{CE}$		"	"
			31					7.0 V						"					"	U/D		"	"
			32									7.0 V		5.5 V					"	P_3		"	"
			33										7.0 V	5.5 V					"	P_2		"	"
			34											7.0 V					"	$\overline{PL}$		"	"
			35														7.0 V		"	CP		"	"
			36											5.5 V				7.0 V	"	P_0		"	"
	I_{IL1}	3009	37	0.5 V										0.0 V					"	P_1	3/	3/	mA
			38					0.5 V						"					"	U/D	"	"	"
			39									0.5 V		0.0 V					"	P_3	"	"	"
			40										0.5 V	0.0 V					"	P_2	"	"	"
			41											0.5 V					"	$\overline{PL}$	"	"	"
			42														0.5 V		"	CP	"	"	"
			43											0.0 V				0.5 V	"	P_0	"	"	"
	I_{IL2}	3009	44				0.5 V				"								"	$\overline{CE}$	"	"	"
	I_{OS}	3011	45	5.5 V	0.0 V		5.5 V							0.0 V						Q_1	-60	-150	
			46				"							"						Q_0	"	"	
			47				"							"						Q_2	"	"	
			48			0.0 V			0.0 V	0.0 V		5.5 V	5.5 V	"	0.0 V	0.0 V	5.5 V	5.5 V		Q_3	"	"	
			49	5.5 V			0.0 V	0.0 V				"	"	"						$\overline{TC}$	"	"	
			50	5.5 V			0.0 V	0.0 V				"	"	"			5.5 V	5.5 V		$\overline{RC}$	"	"	

at end of table III.

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TABLE III. Group A inspection for device 03 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E_1, E_2, X_1, X_2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
				1	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	P_1	Q_1	Q_0	$\overline{CT}$	U/D	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	TC	$\overline{RC}$	CP	P_0	V_{CC}				
1 T_C = +25°C	I_{CC}	3005	51	0.0 V			0.0 V	0.0 V			GND	0.0 V	0.0 V	0.0 V			0.0 V	0.0 V	5.5 V	V_{CC}		55	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = +25°C	Functional 5/	3014	52	B	L	L	B	B	L	L	GND	B	B	B	L	H	A	B	6/	All outputs			
			53	A	H	H	"	"	H	H	"	A	A	B	H	"	"	A	"				
			54	A	H	H	"	"	H	H	"	A	A	A	H	"	"	A	"				
			55	B	L	L	"	"	L	L	"	B	B	B	L	"	"	B	"				
			56	B	L	L	"	"	L	L	"	B	B	A	L	"	"	B	"				
			57	A	H	H	"	"	L	L	"	A	A	A	L	"	2/	A	"				
			58	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			59	"	H	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			60	"	L	L	"	"	H	"	"	"	"	"	"	"	"	"	"				
			61	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			62	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			63	"	H	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			64	"	L	L	"	"	L	H	"	"	"	"	"	"	"	"	"				
			65	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			66	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			67	"	H	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			68	"	L	L	"	"	H	"	"	"	"	"	"	"	"	"	"				
			69	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			70	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			71	"	H	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			72	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"				
			73	"	L	L	"	"	L	L	"	"	"	"	H	L	B	"	"				
			74	"	L	H	"	"	"	"	"	"	"	"	L	H	A	"	"				
			75	"	H	L	"	"	"	"	"	"	"	"	"	"	2/	"	"				
			76	"	H	L	A	"	"	"	"	"	"	"	"	"	"	"	"				
			77	"	H	H	B	A	"	"	"	"	"	"	"	"	"	"	"				
			78	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			79	"	"	"	"	"	"	"	"	"	"	"	H	"	"	"	"				
			80	"	"	"	A	"	"	"	"	"	"	"	"	L	B	"	"				
			81	"	"	"	A	"	"	"	"	"	"	"	"	H	B	"	A				
			82	"	"	"	B	"	"	"	"	"	"	"	"	H	A	"	"				
			83	"	H	H	"	"	H	H	"	"	"	"	"	L	B	"	"				
			84	"	H	L	"	"	"	"	"	"	"	"	"	H	A	"	"				
			85	"	L	H	"	"	"	"	"	"	"	"	"	"	2/	"	"				
			86	"	L	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			87	"	H	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			88	"	H	L	"	"	L	"	"	"	"	"	"	"	"	"	"				
			89	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			90	"	L	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			91	"	H	H	"	"	H	L	"	"	"	"	"	"	"	"	"				
			92	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			93	"	L	H	"	"	"	"	"	"	"	"	"	"	"	"	"				
			94	"	L	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			95	"	H	H	"	"	L	"	"	"	"	"	"	"	"	"	"				
			96	"	H	L	"	"	"	"	"	"	"	"	"	"	"	"	"				
			97	B	L	H	"	"	B	"	"	B	B	B	"	"	A	B	"				
			98	B	L	L	"	"	B	"	"	B	B	B	"	"	B	B	"				
8	Repeat subgroup 7 at T_C = 125°C and T_C = -55°C.																						

See footnotes at end of table III.

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TABLE III. Group A inspection for device 03 - Continued.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases K,L	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases Y,Z 1/	2	3	4	6	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	P_1	Q_1	Q_0	$\overline{CE}$	$\overline{CTD}$	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	TC	$\overline{RC}$	CP	P_0	V_{CC}				
9 T_C = +25°C See fig. 5	f_{MAX} 1/	3003	99 100 101 102		OUT	OUT	GND " "	GND " "	OUT	OUT	GND " "			2.7 V " "			IN " "		5.0 V " "	Q_0 Q_1 Q_2 Q_3	90 " "		MHz " "
	t_{PLH1}	3003	103 104 105 106		OUT	OUT			OUT	OUT	" " " "			2/ " "			" " "		" " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	3.0 " "	7.5 " "	ns " "
	t_{PHL1}	3003	107 108 109 110		OUT	OUT			OUT	OUT	" " " "			" " "			" " "		" " "	CP to Q_0 CP to Q_1 CP to Q_2 CP to Q_3	5.0 " "	11.0 " "	" " "
	t_{PLH2}	3003	111	2.7 V							"	2.7 V	2.7 V	"	OUT		"	2.7 V	"	CP to TC	6.0	13.0	"
	t_{PHL2}	3003	112	"							"	"	"	"	OUT		"		"	CP to TC	5.0	11.0	"
	t_{PLH3}	3003	113	"							"	"	"	"		OUT	"	2.7 V	"	CP to $\overline{RC}$	3.0	7.5	"
	t_{PHL3}	3003	114	"							"	"	"	"		OUT	"		"	CP to $\overline{RC}$	"	7.0	"
	t_{PLH4}	3003	115 116 117 118	" IN 2.7 V "	OUT	OUT	2.7 V " "	2.7 V " "	OUT	OUT	" " " "	" " " IN	" " IN 2.7 V	" " " "			2.7 V " "	IN 2.7 V "	" " "	P_0 to Q_0 P_1 to Q_1 P_2 to Q_2 P_3 to Q_3	" " "	7.0 " "	" " "
	t_{PHL4}	3003	119 120 121 122	" IN 2.7 V "	OUT	OUT	" " "	" " "	OUT	OUT	" " " "	2.7 V " " IN	" " IN 2.7 V	" " " "			" " "	IN 2.7 V "	" " "	P_0 to Q_0 P_1 to Q_1 P_2 to Q_2 P_3 to Q_3	6.0 " "	13.0 " "	" " "
	t_{PLH5}	3003	123	"			IN				"	2.7 V	"	"		OUT		"	"	$\overline{CE}$ to $\overline{RC}$	3.0	7.0	ns
	t_{PHL5}	3003	124	"			IN				"	"	"	"		OUT		"	"	$\overline{CE}$ to $\overline{RC}$	3.0	7.0	"
	t_{PLH6}	3003	125 126 127 128	" 2/ 2.7 V "	OUT	OUT	2.7 V " "	2.7 V " "	OUT	OUT	" " " "	" " " 2/	" " 2/ 2.7 V	IN " "			2.7 V " "	2/ 2.7 V "	" " "	$\overline{PL}$ to Q_0 $\overline{PL}$ to Q_1 $\overline{PL}$ to Q_2 $\overline{PL}$ to Q_3	5.0 " "	11.0 " "	" " "
	t_{PHL6}	3003	129 130 131 132	" 0.0 V 2.7 V 2.7 V	OUT	OUT	" " "	" " "	OUT	OUT	" " " "	2.7 V " " 0.0 V	" " 0.0 V 2.7 V	" " " "			" " "	0.0 V 2.7 V "	" " "	$\overline{PL}$ to Q_0 $\overline{PL}$ to Q_1 $\overline{PL}$ to Q_2 $\overline{PL}$ to Q_3	5.5 " "	12.0 " "	" " "

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TABLE III. Group A inspection for device 03 - Cont

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases X,2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	P_1	Q_1	Q_0	$\overline{CE}$	$\overline{U}/D$	Q_2	Q_3	GND	P_3	P_2	$\overline{PE}$	TC	$\overline{RC}$	CP	P_0	V_{CC}				
9 T_C = +25°C See fig. 5	t_{PLH7}	3003	133					IN			GND			2/		OUT			5.0 V	$\overline{U}/D$ to $\overline{RC}$	7.0	18.0	ns
	t_{PHL7}	3003	134					"			"			"		OUT			"	$\overline{U}/D$ to $\overline{RC}$	5.5	12.0	"
	t_{PLH8}	3003	135					"			"			"	OUT				"	$\overline{U}/D$ to TC	4.0	10.0	"
	t_{PHL8}	3003	136					"			"			"	OUT				"	$\overline{U}/D$ to TC	4.0	10.0	"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 10, except T_C = -55°C.																						

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TABLE III. Group A inspection for device 04.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F Cases X,2 1/	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	P_1	Q_1	Q_0	CPB	CPU	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	TCU	TCD	$\overline{MR}$	P_0	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6	2.0 V 5.5 V 0.0 V	-1.0 mA	-1.0 mA	5.5 V 2.0 V	2.0 V 5.5 V	-1.0 mA	-1.0 mA	GND 	2.0 V 5.5 V 0.0 V	2.0 V 5.5 V 0.0 V	0.8 V 	-1.0 mA	-1.0 mA	0.8 V 	2.0 V 5.5 V 0.0 V	4.5 V 	Q_1 Q_0 Q_2 Q_3 TCU TCD	2.5 		V
	V_{OL}	3007	7 8 9 10 11 12	0.8 V 5.5 V 0.0 V	20 mA	20 mA	5.5 V 0.8 V	0.8 V 5.5 V	20 mA	20 mA	 	0.8 V 5.5 V 0.0 V	0.8 V 5.5 V 0.0 V	 	20 mA	20 mA	0.0 V 0.8 V 0.0 V	0.8 V 5.5 V 0.0 V	 	Q_1 Q_0 Q_2 Q_3 TCU TCD		0.5 	
	V_{IC}	3022	13 14 15 16 17 18 19 20	-18 mA			-18 mA	-18 mA			 	-18 mA	-18 mA	-18 mA			-18 mA	-18 mA	 	P_1 CPD CPU P_3 P_2 PL MR P_0		-1.2 	
	I_{IH1}	3010	21 22 23 24 25 26 27 28	2.7 V 			2.7 V	2.7 V			 	2.7 V	2.7 V	2.7 V			2.7 V	2.7 V	5.5 V 	P_1 CPD CPU P_3 P_2 PL MR P_0		20 	μA
	I_{IH2}	3010	29 30 31 32 33 34 35 36	7.0 V 			7.0 V	7.0 V			 	7.0 V	7.0 V	7.0 V			7.0 V	7.0 V	 	P_1 CPD CPU P_3 P_2 PL MR P_0		100 	
	I_{IL1}	3009	37 38 39 40 41 42	0.5 V 							 	0.5 V	0.5 V	0.0 V 0.5 V 0.0 V			0.0 V 0.5 V 0.0 V	0.5 V	 	P_1 P_3 P_2 PL MR P_0	3/ 	3/ 	mA
	I_{IL2}	3009	43 44				0.5 V 5.5 V	5.5 V 0.5 V			 								 	CPD CPU	 	 	
	I_{OS}	3011	45 46 47 48 49 50	5.5 V 0.0 V	0.0 V	0.0 V	 5.5 V	 5.5 V	0.0 V	0.0 V	 	5.5 V 5.5 V 0.0 V	5.5 V 0.0 V	0.0 V 	0.0 V	0.0 V	0.0 V 	5.5 V 5.5 V 0.0 V	 	Q_1 Q_0 Q_2 Q_3 TCU TCD	-60 	-150 	
	I_{CC}	3005	51	5.5 V			5.5 V	5.5 V			"	5.5 V	5.5 V	"			"	5.5 V	5.5 V	V_{CC}		55	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						

See footnotes at end of table III.

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases X,2 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	P_1	Q_1	Q_0	CPD	CPU	Q_2	Q_3	GND	P_3	P_2	$\overline{PL}$	TCU	TCD	MR	P_0	V_{CC}				
7 T_C = +25°C	Functional 5/	3014	52	A	L	L	A	A	L	L	GND	A	A	A	H	H	A	A	6/	All outputs			
			53		H	H			H	H				B			B	B					
			54											A									
			55	B	L	L		B	L	L		B	B	A	L	L		B					
			56			H		A						B									
			57		H	L		B	L	L				A			B						
			58			H		A															
			59			L		B															
			60			H		A															
			61		H	L		B	H	H													
			62			L		A															
			63			H		B															
			64			L		A															
			65		L	H		B	L	L													
			66			L		A															
			67			H		B															
			68			H		A															
			69		H	L		B	H	H													
			70			L		A															
			71			H		B															
			72			H		A															
			73		L	L		B	L	L													
			74	A		L		A				A	A					A					
			75			H		B															
			76			H		A															
			77			L		B															
			78			H		A															
			79		H	L		B	H	H													
			80			L		A															
			81			H		B															
			82			H		A															
			83		L	L		B	L	L													
			84			H		A															
			85			L		B															
			86			H		A															
			87			L		B															
			88		L	L		A	L	L													
			89			H		B															
			90			L		A															
			91			H		B															
			92			H		A															
			93		L	L		B	L	L													
			94			H		A															
			95			L		B															
			96			H		A															
			97			L		B															
			98			H		A															
			99		L	L		B	L	L													
			100			H		A															
			101			L		B															
			102			H		A															
			103			L		B															
			104			H		A															
			105			H		B															
			106	B		L		A				B	B					B					
			107			H		B	H	L													
			108			L		A															
			109			H		B															
			110			H		A															
			111			L		B															
			112			H		A															
			113			L		A															

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / P_1	2 / 3 / Q_1	3 / 4 / Q_0	4 / 5 / CPD	5 / 7 / CPU	6 / 8 / Q_2	7 / 9 / Q_3	8 / 10 / GND	9 / 12 / P_3	10 / 13 / P_2	11 / 14 / PE	12 / 15 / TCU	13 / 17 / TCD	14 / 18 / MR	15 / 19 / P_0	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 T_C = +25°C	Functional 5/	3014	114 115 116 117 118 119 120 121	B " " " " " " "	L H " " " L " "	L H H L L H H L	B A B A B A B A	A " " " " " " "	H L " " " " " "	L " " " " " " "	GND " " " " " " "	B " " " " " " "	B " " " " " " "	A " " " " " " "	H " " " " " " "	H " " " " " " "	B " " " " " " "	B " " " " " " "	6/ " " " " " " "	All outputs " " " " " " "			
8	Repeat subgroup 7 at T_C = +125°C and T_C = -55°C.																						
9 T_C = +25°C See fig. 5	f_{MAX} 7/	3003	122 123 124 125 126 127 128 129		OUT OUT 	OUT OUT 	2.7 V " " IN " " 	IN " " 2.7 V " " 	OUT OUT 	OUT OUT 	GND " " " " " " "			2.7 V " " " " " " "			GND " " " " " " "		5.0 V " " " " " " "	CPU to Q_0 CPU to Q_1 CPU to Q_2 CPU to Q_3 CPD to Q_0 CPD to Q_1 CPD to Q_2 CPD to Q_3	90 " " " " " " "		MHz " " " " " " "
	t_{PLH1}	3003	130 131 132 133 134 135 136 137	0.0 V 0.0 V 0.0 V " "	OUT OUT 	OUT OUT 	 IN " " 	IN " " 	OUT OUT 	OUT OUT 	" " " " " " " "	0.0 V " " " " " 0.0 V "	0.0 V " 0.0 V 0.0 V 0.0 V	2/ " " " " " " "			0.0 V " " " " " " "	0.0 V 0.0 V " 	" " " " " " " "	CPU to Q_0 CPU to Q_1 CPU to Q_2 CPU to Q_3 CPD to Q_0 CPD to Q_1 CPD to Q_2 CPD to Q_3	3.5 " " " " " " "	8.5 " " " " " " "	ns " " " " " " "
	t_{PHL1}	3003	138 139 140 141 142 143 144 145	0.0 V 0.0 V 0.0 V 0.0 V	OUT OUT 	OUT OUT 	2.7 V " " IN " " 	IN " " 2.7 V " " 	OUT OUT 	OUT OUT 	" " " " " " " "	" " " " " " " "	" " " 0.0 V 0.0 V 0.0 V	" " " " " " " "			" " " " " " " "	 0.0 V " 	" " " " " " " "	CPU to Q_0 CPU to Q_1 CPU to Q_2 CPU to Q_3 CPD to Q_0 CPD to Q_1 CPD to Q_2 CPD to Q_3	5.5 " " " " " " "	12.5 " " " " " " "	" " " " " " " "
	t_{PLH2}	3003	146					IN			"			"	OUT		"			"	CPU to TCU	4.0	9.0	"
	t_{PHL2}	3003	147				2.7 V	IN			"			"	OUT		"	0.0 V		"	CPU to TCU	3.5	8.0	"
	t_{PLH3}	3003	148 149 150 151	0.0 V 2.7 V 0.0 V 0.0 V	OUT	OUT	0.0 V " " 	0.0 V " " 	OUT	OUT	" " " "	0.0 V " " 2.7 V	0.0 V 0.0 V 2.7 V 0.0 V	IN " " 				2.7 V 0.0 V " 	" " " "	PE to Q_0 PE to Q_1 PE to Q_2 PE to Q_3	5.0 " " 	11.0 " " 	" " " "

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / P_1	2 / 3 / Q_1	3 / 4 / Q_0	4 / 5 / CPD	5 / 7 / CPU	6 / 8 / Q_2	7 / 9 / Q_3	8 / 10 / GND	9 / 12 / P_3	10 / 13 / P_2	11 / 14 / $\overline{PL}$	12 / 15 / TCU	13 / 17 / TCD	14 / 18 / MR	15 / 19 / P_0	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C See fig. 5	t_{PHL3}	3003	152	0.0 V	OUT	OUT	0.0 V	0.0 V			GND	0.0 V	0.0 V	IN			0.0 V	0.0 V	5.0 V	$\overline{PL}$ to Q_0	5.5	13.0	ns
			153	"			"	"			"	"	"	"			"	"	"	$\overline{PL}$ to Q_1	"	"	"
			154	"			"	"	OUT		"	"	"	"			"	"	"	$\overline{PL}$ to Q_2	"	"	"
			155	"			"	"		OUT	"	"	"	"			"	"	"	$\overline{PL}$ to Q_3	"	"	"
	t_{PLH4}	3003	156	"			IN				"	"	"	2/		OUT	"	"	"	CPD to TCD	4.0	9.0	"
	t_{PHL4}	3003	157	"			IN				"	"	"	2/		OUT	"		"	CPD to TCD	3.5	8.0	"
	t_{PLH5}	3003	158	IN	OUT	OUT	0.0 V	0.0 V			"			0.0 V			"	IN	"	P_0 to Q_0	3.0	7.0	"
			159				"	"			"			"			"		"	P_1 to Q_1	"	"	"
			160				"	"	OUT		"		IN	"			"		"	P_2 to Q_2	"	"	"
			161				"	"		OUT	"	IN		"			"		"	P_3 to Q_3	"	"	"
	t_{PHL5}	3003	162	IN	OUT	OUT	"	"			"			"			"	IN	"	P_0 to Q_0	6.0	14.5	"
			163				"	"			"			"			"		"	P_1 to Q_1	"	"	"
			164				"	"	OUT		"		IN	"			"		"	P_2 to Q_2	"	"	"
			165				"	"		OUT	"	IN		"			"		"	P_3 to Q_3	"	"	"
	t_{PLH6}	3003	166					"			"			2/	OUT		IN		"	MR to TCU	5.0	13.5	"
	t_{PHL6}	3003	167				0.0 V	2.7 V			"			2/		OUT	IN		"	MR to TCD	6.0	14.5	"
	t_{PLH7}	3003	168					0.0 V			"			IN	OUT		0.0 V	0.0 V	"	$\overline{PL}$ to TCU	7.0	15.5	"
	t_{PLH8}	3003	169				0.0 V				"			"		OUT			"	$\overline{PL}$ to TCD	"	15.5	"
	t_{PHL7}	3003	170				2.7 V	0.0 V			"			"	OUT				"	$\overline{PL}$ to TCU	"	14.5	"
	t_{PHL8}	3003	171	0.0 V			0.0 V	2.7 V			"	0.0 V	0.0 V	"		OUT	0.0 V	0.0 V	"	$\overline{PL}$ to TCD	6.0	14.5	"
	t_{PLH9}	3003	172	IN				0.0 V			"			0.0 V	OUT		"	IN	5.0 V	P_0 to TCU	7.0	14.5	"
			173					"			"			"	"		"		"	P_1 to TCU	"	"	"
			174					"			"		IN	"	"		"		"	P_2 to TCU	"	"	"
			175					"			"	IN		"	"		"		"	P_3 to TCU	"	"	"
	t_{PHL9}	3003	176	IN				"			"			"	"		"	IN	5.0 V	P_0 to TCU	5.0	14.0	"
			177					"			"			"	"		"		"	P_1 to TCU	"	"	"
			178					"			"		IN	"	"		"		"	P_2 to TCU	"	"	"
			179					"			"	IN		"	"		"		"	P_3 to TCU	"	"	"

See footnotes at end of table III.

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TABLE III. Group A inspection for device 04 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

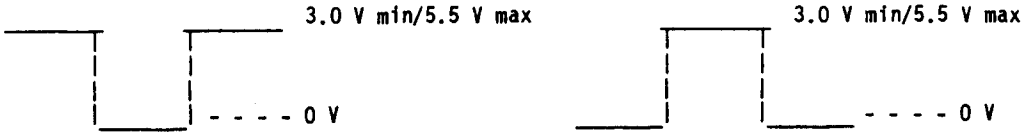
Subgroup	Symbol	MIL-STD-883 method	Cases E,F / Cases X,2 1/ / Test no.	1 / 2 / P_1	2 / 3 / Q_1	3 / 4 / Q_0	4 / 5 / CPD	5 / 7 / CPU	6 / 8 / Q_2	7 / 9 / Q_3	8 / 10 / GND	9 / 12 / P_3	10 / 13 / P_2	11 / 14 / $\overline{PL}$	12 / 15 / $\overline{TCU}$	13 / 17 / $\overline{TCD}$	14 / 18 / MR	15 / 19 / P_0	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C See fig. 5	t_{PLH10}	3003	180	0.0 V			0.0 V	2.7 V			GND	0.0 V	0.0 V	0.0 V		OUT	0.0 V	IN	5.0 V	P_0 to $\overline{TCD}$	5.5	14.5	ns
			181	IN			"	"			"	"	0.0 V	"		"	"	0.0 V	"	P_1 to $\overline{TCD}$	"	"	"
			182	0.0 V			"	"			"	"	IN	"		"	"	"	"	P_2 to $\overline{TCD}$	"	"	"
			183	"			"	"			"	IN	0.0 V	"		"	"	"	"	P_3 to $\overline{TCD}$	"	"	"
	t_{PHL10}	3003	184	"			"	"			"	0.0 V	"	"		"	"	IN	"	P_0 to $\overline{TCD}$	6.5	14.0	"
			185	IN			"	"			"	"	"	"		"	"	0.0 V	"	P_1 to $\overline{TCD}$	"	"	"
			186	0.0 V			"	"			"	"	IN	"		"	"	"	"	P_2 to $\overline{TCD}$	"	"	"
			187	"			"	"			"	IN	0.0 V	"		"	"	"	"	P_3 to $\overline{TCD}$	"	"	"
	t_{PHL11}	3003	188	"			"	0.0 V			"	0.0 V	"	2/			IN	2.7 V	"	MR to Q_0	5.0	14.5	"
			189	2.7 V	OUT	OUT	"	"			"	"	"	"			"	0.0 V	"	MR to Q_1	"	"	"
			190	0.0 V			"	"	OUT		"	"	2.7 V	"			"	"	"	MR to Q_2	"	"	"
			191	0.0 V			"	"		OUT	"	2.7 V	0.0 V	"			"	"	"	MR to Q_3	"	"	"
10	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and use limits from table I.																						
11	Same tests and terminal conditions as subgroup 10, except T_C = -55°C.																						

See footnotes at end of table III.

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1/ For cases X and 2, pins not referenced are NC.

2/ Apply one pulse prior to measurement.



3/

Parameter	Device	Circuit A	Circuit B	Circuit C	Circuit D
I_{IL1}	All	-.25/-0.6	-.03/-0.6	-.25/-0.6	0.0/-0.6
I_{IL2}	01, 02	-.50/-1.2	-.06/-1.2	-.50/-1.2	0.0/-0.6
I_{IL2}	03, 04	-.75/-1.8	-.09/-1.8	-.50/-1.8	

4/ For types 01 and 02, set outputs to 15th count (P_0, P_1, P_2, P_3 = 1), prior to measurement.

5/ H $\geq$ 1.5 V, L $\leq$ 1.5 V, A = 3.0 V minimum; B = 0.0 V or GND.

6/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

7/ The f_{MAX} minimum limit specified is the frequency of the input pulse. The output frequency shall be one-half of the input frequency.

8/ For types 01 and 02, increment such that measurement of the specified output can occur on the next applied CP.

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4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete Part or Identifying Number (PIN) (see 6.7).

d. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

e. Requirements for certificate of compliance, if applicable.

f. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

g. Requirements for failure analysis (including required test condition of method 5005 of MIL-STD-883), corrective action, and reporting of results, if applicable.

h. Requirements for product assurance options.

i. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the PIN. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

j. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - Ground zero voltage potential.

I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal.

V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the PIN.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F161A
02	54F163A
03	54F191
04	54F193

6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' desgnations.

Device type	Circuit			
	A	B	C	D
	National	Motorola	Signetics	TI
01	X		X	X
02	X		X	X
03	X			
04	X			

6.7 Part or Identifying Number (PIN). The PIN is created as specified in MIL-M-38510.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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